Electric Vehicle Service Manual

PREVENTIVE MAINTENANCE GUIDE

Author: Generative AI

Demo Purpose

Table of Contents

INTRODUCTION **3**

Purpose and Scope 3
Why Preventive Maintenance Matters for EVs *4*
How to Use This Manual *4*

SECTION 1: BATTERY SYSTEM MONITORING **4**

Battery Failure Prediction Scenarios 5
1.1 Cell Voltage Imbalance 5
1.2 Battery Thermal Runaway Risk 5
1.3 Accelerated Capacity Degradation 6
1.4 Battery Internal Resistance Increase 7
1.5 BMS Communication Failure 8

SECTION 2: ELECTRIC MOTOR SYSTEM MONITORING **9**

Motor Failure Prediction Scenarios 9
2.1 Motor Overheating (Thermal Stress) 10
2.2 Bearing Wear or Lubrication Failure 11
2.3 Rotor Eccentricity / Misalignment 12
2.4 Stator Winding Short / Insulation Breakdown 13
2.5 Demagnetization of Permanent Magnets (PMSM Motors) 14
2.6 Phase Imbalance or Open Circuit 15
2.7 Cooling System Degradation 16

SECTION 3: POWER ELECTRONICS (INVERTER) MONITORING **17**

Inverter Failure Prediction Scenarios 17
3.1 Inverter Overheating (Thermal Overload) 17
3.2 DC Bus Overvoltage / Undervoltage 18
3.3 Phase Current Imbalance 19
3.4 IGBT/MOSFET Degradation (Switching Failure) 20
3.5 DC Link Capacitor Aging 21
3.6 Insulation Breakdown / Ground Fault 22
3.7 Connector Corrosion or Contact Resistance 23

SECTION 4: ON-BOARD CHARGER (OBC) MONITORING **24**

OBC Failure Prediction Scenarios 24
4.1 Charging Rate Degradation 24
4.2 OBC Overheating 25
4.3 Power Factor and Efficiency Degradation 26
4.4 Communication and Control Faults 27
4.5 Input Voltage Instability 28

SECTION 5: TIRE SYSTEM MONITORING **29**

Tire Failure Prediction Scenarios 30
5.1 Rapid Air Pressure Loss (Puncture/Blowout) 30
5.2 Slow Pressure Leak 31
5.3 Chronic Under-Inflation 32

5.4 Over-Inflation 33
5.5 Excessive Heat Build-Up 34
5.6 Uneven Tire Wear (Alignment/Suspension Issues) 35
5.7 Sidewall Damage (Curb/Pothole Impact) 36
5.8 Tire Imbalance or Structural Deformation 37
5.9 Tread Separation or Delamination 38
5.10 Valve Stem or Rim Leak 39
5.11 Tire Aging (Rubber Degradation/Cracking) 40
5.12 Overload or High Payload Operation 41
5.13 Seasonal or Temperature-Induced Pressure Fluctuations 42
5.14 Combined Predictive Scenario: Multi-Factor Tire Risk Index 43

SECTION 6: ENERGY MANAGEMENT SYSTEM MONITORING 44

Energy Management Failure Prediction Scenarios 44
6.1 Power Distribution Inefficiency 44
6.2 Regenerative Braking Degradation 45
6.3 Range Prediction Accuracy Degradation 46
6.4 Auxiliary System Power Drain 47

SECTION 7: THERMAL MANAGEMENT SYSTEM MONITORING 49

Thermal Management Failure Prediction Scenarios 49
7.1 Cooling System Pump Failure 49
7.2 Radiator or Heat Exchanger Blockage 50
7.3 Coolant Leak or Low Fluid Level 51
7.4 Coolant Quality Degradation 52
7.5 Thermal Control Valve Malfunction 53
7.6 Battery Thermal Management Failure 54

SECTION 8: HIGH VOLTAGE SAFETY SYSTEM MONITORING 55

High Voltage Safety Failure Prediction Scenarios 56
8.1 Insulation Resistance Degradation 56
8.2 Ground Fault Detection 57
8.3 Interlock System Failure 58
8.4 HV Contactor Degradation 59
8.5 Arc Fault Detection Events 60

SECTION 9: COMPREHENSIVE MONITORING SCHEDULE 61

Daily Checks (Before Each Drive) 61
Weekly Checks (5-10 minutes) 62
Monthly Checks (15-20 minutes) 62
Quarterly Checks (30-45 minutes) 63
Annual Checks (Professional Service Recommended) 64

SECTION 10: EMERGENCY PROCEDURES 65

10.1 Battery Thermal Event Response 65
IMMEDIATE ACTIONS (Do not delay): 65
10.2 High Voltage Exposure Protocol 66
COLLISION WITH HV SYSTEM DAMAGE: 67
10.3 Vehicle Immobilization and Securing 68

10.3.1 Standard Immobilization Procedure *68*
10.3.2 Emergency Immobilization (Fire or Electrical Hazard) *69*
10.3.3 Collision Immobilization Procedure *70*
10.3.4 Towing and Recovery Procedures *71*
10.3.5 Long-Term Storage Immobilization *72*
10.4 Emergency Services Communication 73
10.4.1 Initial Emergency Call Information *73*
10.4.2 Emergency Responder Information Sheet *73*
10.4.3 Fire Department Specific Information *74*
10.4.4 Extrication and Rescue Information *75*
10.4.5 Post-Incident Monitoring and Recovery *76*

SECTION 11: COMPREHENSIVE MONITORING SCHEDULE 77

11.1 Daily Checks (Every Drive) 77
11.2 Weekly Checks (Every 7 Days or 500 km) 78
11.3 Monthly Checks (Every 30 Days or 2,000 km) 79
11.4 Quarterly Checks (Every 3 Months or 6,000 km) 80
11.5 Annual Checks (Every 12 Months or 20,000 km) 83

SECTION 12: GLOSSARY OF TECHNICAL TERMS 84

SECTION 13: QUICK REFERENCE EMERGENCY GUIDE 85

⚠ CRITICAL WARNINGS - IMMEDIATE ACTION REQUIRED 85
Emergency Contact Numbers 86
Quick Diagnostic Reference 86
Emergency Shutdown Procedure 87
Thermal Event Response (Battery Fire) 88
Collision Response Checklist 88
Towing Quick Reference 89
Pressure and Voltage Quick Reference 89
Service Timing Quick Reference 90
Owner-Serviceable vs. Professional Service 91
Emergency Equipment Recommended 92

INTRODUCTION

Purpose and Scope

This comprehensive service manual is designed for electric vehicle fleet operators, maintenance supervisors, and drivers who need to monitor vehicle health and identify potential issues before they result in breakdowns or costly repairs. By leveraging cloud-connected vehicle data and systematic monitoring protocols, you can transition from reactive repairs to predictive maintenance.

Why Preventive Maintenance Matters for EVs

Electric vehicles contain sophisticated high-voltage systems, precision electronics, and thermal management components that require different maintenance approaches than conventional vehicles. While EVs have fewer moving parts, the criticality of their electronic systems means that early detection of degradation is essential for:

- **Maximizing uptime**: Preventing unexpected breakdowns that sideline vehicles
- **Extending component life**: Addressing minor issues before they cascade into major failures
- **Ensuring safety**: Identifying high-voltage system degradation before hazardous conditions develop
- **Optimizing total cost of ownership**: Reducing emergency repairs and maximizing component lifespan
- **Maintaining performance**: Preserving range, efficiency, and driver satisfaction

How to Use This Manual

This manual is structured to support both:

1. **Systematic monitoring**: Regular checks using cloud data dashboards and vehicle telematics
2. **Issue identification**: Recognizing early warning signs through data trends and physical symptoms
3. **Preventive action**: Taking appropriate measures before scheduled service appointments

Each system section includes:

- Component descriptions and failure modes
- Exact threshold values for normal, warning, and critical conditions
- Cloud data inputs required for prediction
- Preventive suggestions for drivers and fleet managers
- Service timing recommendations

⚠️ **SAFETY FIRST**: This manual covers monitoring and basic preventive measures only. High-voltage system repairs must be performed by certified technicians with appropriate safety equipment and training.

SECTION 1: BATTERY SYSTEM MONITORING

The high-voltage battery pack is the energy storage heart of your EV, typically consisting of:

- **Battery cells**: Individual lithium-ion cells (hundreds to thousands per pack)
- **Modules**: Groups of cells connected in series/parallel configurations
- **Battery Management System (BMS)**: Electronic controller monitoring voltage, temperature, and state of charge
- **Thermal management**: Cooling/heating system maintaining optimal temperature
- **Contactors and fuses**: High-voltage disconnect and protection devices

Battery Failure Prediction Scenarios

1.1 Cell Voltage Imbalance

Description: Individual cells within the battery pack develop different voltage levels, indicating degradation, manufacturing defects, or BMS calibration issues.

Data Inputs Required:

- Individual cell voltages (all cells in pack)
- Pack voltage
- State of Charge (SOC)
- Charging/discharging current
- Cell temperature distribution

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Cell voltage deviation	<30mV between cells	30-75mV	>75mV
Module voltage deviation	<100mV between modules	100-200mV	>200mV
Voltage spread at 50% SOC	<50mV	50-100mV	>100mV

Early Warning Signs:

- Gradual increase in cell voltage spread over weeks/months
- Specific cells consistently at voltage extremes (highest or lowest)
- Voltage imbalance worsening at high SOC (>80%)
- BMS performing frequent cell balancing operations

Preventive Suggestions for Drivers:

- Perform monthly "balancing charge": charge to 100% and hold for 2-3 hours
- Avoid frequent charging to 100% or discharging below 10% (causes stress)
- Monitor charging completion time—significant increases indicate balancing issues
- Park in moderate temperatures when possible (15-25°C / 59-77°F)

When Service is Required:

- **Immediate**: Voltage deviation >150mV, "Battery Fault" warning, sudden range loss >20%
- **Within 1 week**: Voltage deviation 75-150mV, charging time increased >25%
- **Next scheduled service**: Voltage deviation 30-75mV, gradual trend worsening

1.2 Battery Thermal Runaway Risk

Description: Excessive heat generation in battery cells due to internal short circuits, overcharging, or

cooling system failure. Can lead to catastrophic thermal events.

Data Inputs Required:

- Cell temperature sensors (multiple points across pack)
- Ambient temperature
- Coolant inlet/outlet temperature
- Cooling pump speed and flow rate
- Charging current and voltage
- Recent fast-charging history

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Cell temperature	15-35°C (59-95°F)	35-45°C (95-113°F)	>45°C (>113°F)
Cell temp deviation	<5°C between cells	5-10°C	>10°C
Temp rise during charging	<15°C above ambient	15-25°C	>25°C
Coolant ΔT (in-out)	3-8°C	8-15°C	>15°C

Early Warning Signs:

- One module consistently 5-8°C hotter than others
- Battery temperature remaining elevated (>35°C) 30+ minutes after driving
- Cooling fan running continuously or at high speed
- Temperature spikes during fast charging or high-power acceleration
- "Battery Temperature High" warnings during moderate driving

Preventive Suggestions for Drivers:

- Pre-condition battery before fast charging (use vehicle app to cool/heat)
- Avoid back-to-back fast charging sessions (allow 30-minute cooldown)
- Limit high-power acceleration in hot weather (>35°C ambient)
- Park in shade during hot weather
- Monitor cooling system warnings and address immediately
- Avoid charging immediately after high-speed highway driving

When Service is Required:

- **STOP IMMEDIATELY**: Battery temp >50°C, smoke/burning smell, rapid temperature rise
- **Immediate**: Persistent temp >45°C, cooling fan failure, cell temp deviation >12°C
- **Within 1 week**: Frequent temp warnings, cell temp deviation 8-12°C, coolant level low
- **Next scheduled service**: Gradual temp increase trend, cell deviation 5-8°C

1.3 Accelerated Capacity Degradation

Description: Battery capacity (total energy storage) decreases faster than normal aging rates, reducing vehicle range and usable energy.

Data Inputs Required:

- State of Health (SOH) percentage
- Full charge capacity (kWh)
- Energy consumed per charge cycle
- Total charge cycles
- Fast charge vs. Level 2 charge ratio
- Average SOC operating range
- Temperature exposure history

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Capacity degradation rate	<3% per year	3-5% per year	>5% per year
State of Health (SOH)	>90%	85-90%	<85%
Range loss vs. new	<10%	10-20%	>20%
Capacity fade per 1000 cycles	<2%	2-4%	>4%

Early Warning Signs:

- Displayed range at 100% SOC decreasing faster than 2-3% per year
- Actual range significantly less than displayed estimate (>15% difference)
- Rapid SOH decline over 3-6 months
- Vehicle requiring more frequent charging for same usage patterns
- Charging to 100% taking less time (indicates reduced capacity)

Preventive Suggestions for Drivers:

- Maintain SOC between 20-80% for daily driving (reduces stress)
- Limit fast charging to 1-2 times per week maximum
- Avoid extreme temperatures: pre-condition cabin before driving
- Minimize time spent at very high (>90%) or very low (<10%) SOC
- Use "battery protection" or "range mode" features if available
- Avoid leaving vehicle at 100% SOC for extended periods (>24 hours)
- For long-term storage (>2 weeks), maintain 50-60% SOC

When Service is Required:

- **Immediate**: Sudden capacity loss >10% in one month, SOH <80%
- **Within 1 week**: Degradation rate >6% per year, SOH 80-85%
- **Within 1 month**: Degradation rate 4-5% per year, SOH 85-88%
- **Next scheduled service**: Degradation rate 3-4% per year, SOH 88-90%

1.4 Battery Internal Resistance Increase

Description: Internal resistance rises due to electrode degradation, electrolyte decomposition, or contact resistance, reducing power delivery and increasing heat generation.

Data Inputs Required:

- Battery voltage under load vs. no-load
- Current draw during acceleration
- Power delivery vs. SOC correlation
- Temperature rise during discharge
- Charging efficiency (energy in vs. energy stored)

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Voltage sag under load	<5V drop at rated power	5-10V drop	>10V drop
Power delivery	>95% of rated	85-95%	<85%
Charging efficiency	>92%	88-92%	<88%
Heat generation rate	<0.5°C per kWh	0.5-1.0°C per kWh	>1.0°C per kWh

Early Warning Signs:
- Reduced acceleration performance, especially from high SOC
- "Power Limited" messages during normal acceleration
- Battery heating more during charging or discharging
- Voltage dropping more significantly under load
- Regenerative braking power acceptance reduced

Preventive Suggestions for Drivers:
- Avoid sustained high-power demands (extended highway speeds, heavy towing)
- Monitor power delivery performance monthly using consistent acceleration tests
- Maintain optimal operating temperature (avoid extreme cold/hot)
- Reduce load when "Power Limited" warnings appear
- Consider gentler driving style if resistance increasing

When Service is Required:
- **Immediate**: Voltage sag >12V, power delivery <80%, persistent power limiting
- **Within 1 week**: Voltage sag 10-12V, power delivery 80-85%
- **Within 1 month**: Voltage sag 7-10V, power delivery 85-90%
- **Next scheduled service**: Voltage sag 5-7V, charging efficiency 88-92%

1.5 BMS Communication Failure

Description: Battery Management System loses communication with vehicle systems or fails to properly monitor/control battery functions.

Data Inputs Required:
- BMS diagnostic trouble codes (DTCs)
- CAN bus communication errors
- BMS software version
- Sensor data dropout frequency
- Contactor operation logs

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Communication errors	0 per day	1-5 per day	>5 per day
Sensor data gaps	0%	<1% of readings	>1%
BMS resets	0 per month	1-2 per month	>2 per month

Early Warning Signs:

- Intermittent "Battery System Fault" warnings
- SOC display jumping or showing inconsistent values
- Charging sessions failing to start or stopping unexpectedly
- Vehicle refusing to start with adequate battery charge
- Dashboard warnings about battery communication

Preventive Suggestions for Drivers:

- Keep vehicle software updated (OTA updates or dealer service)
- Document all error messages with timestamps and conditions
- Perform "12V battery" check (low 12V can cause BMS communication issues)
- Avoid aftermarket modifications to electrical systems
- Report intermittent warnings even if they clear themselves

When Service is Required:

- **Immediate**: Vehicle won't start, persistent "Battery Fault", charging impossible
- **Within 1 week**: Daily communication errors, frequent BMS resets
- **Within 1 month**: Weekly communication errors, intermittent charging issues
- **Next scheduled service**: Occasional errors, no functional impact

SECTION 2: ELECTRIC MOTOR SYSTEM MONITORING

The electric motor converts electrical energy into mechanical motion. EV motors typically include:

- **Stator**: Stationary electromagnetic windings
- **Rotor**: Rotating component (permanent magnets or induction type)
- **Bearings**: Support rotating shaft with minimal friction
- **Cooling system**: Liquid or air cooling to manage heat
- **Position sensors**: Encoders or resolvers for precise control
- **Shaft and coupling**: Connects motor to drivetrain

Motor Failure Prediction Scenarios

2.1 Motor Overheating (Thermal Stress)

Description: Excessive motor temperature caused by prolonged high torque demands, cooling system degradation, or bearing friction.

Data Inputs Required:

- Motor stator temperature
- Motor rotor temperature (if available)
- Bearing temperature
- Coolant inlet/outlet temperature
- Ambient temperature
- Torque demand history
- Vehicle speed and gradient data
- Cooling pump speed

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Stator temperature	60-90°C (140-194°F)	90-110°C (194-230°F)	>110°C (>230°F)
Rotor temperature	70-100°C (158-212°F)	100-120°C (212-248°F)	>120°C (>248°F)
Bearing temperature	50-80°C (122-176°F)	80-100°C (176-212°F)	>100°C (>212°F)
Coolant ΔT	5-15°C	15-25°C	>25°C
Temp rise rate	<2°C per minute	2-4°C per minute	>4°C per minute

Early Warning Signs:

- Motor temperature rising 8-10°C over weeks for same driving conditions
- Temperature remaining elevated (>80°C) 15+ minutes after stopping
- Cooling fan running continuously during normal driving
- "Motor Temperature High" warnings during moderate acceleration
- Reduced power delivery when motor is hot

Preventive Suggestions for Drivers:

- Avoid sustained high-speed driving (>120 km/h) in hot weather (>35°C)
- Take 10-minute cooling breaks during extended uphill climbs or towing
- Monitor coolant level monthly (check when cold)
- Reduce power demands if temperature warnings appear
- Allow motor to cool before fast charging after spirited driving
- Avoid "lugging" the motor at low speeds with high torque demands
- Use "Eco" or "Efficiency" mode during hot weather operation

When Service is Required:

- **STOP IMMEDIATELY**: Motor temp >125°C, burning smell, smoke, grinding noise
- **Immediate**: Persistent temp >110°C, cooling system leak, bearing noise
- **Within 1 week**: Frequent temp >100°C, coolant ΔT >20°C, temp rise trend
- **Next scheduled service**: Occasional temp 90-100°C, gradual temp increase

2.2 Bearing Wear or Lubrication Failure

Description: Motor bearings degrade due to contamination, inadequate lubrication, misalignment, or normal wear, causing friction, vibration, and eventual seizure.

Data Inputs Required:

- High-frequency vibration data (accelerometer near motor)
- Acoustic noise spectrum (if microphone sensors present)
- Motor torque vs. current ratio
- Bearing temperature
- Motor efficiency (power in vs. mechanical power out)
- Rotational speed variations

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Vibration amplitude	<0.5 g RMS	0.5-1.5 g RMS	>1.5 g RMS
Bearing noise (dB)	<65 dB	65-75 dB	>75 dB
Torque ripple	<3%	3-7%	>7%
Efficiency loss	<2% from baseline	2-5%	>5%
Temperature differential	<10°C (bearing vs. motor)	10-20°C	>20°C

Early Warning Signs:

- New humming, whining, or grinding sounds during acceleration
- Vibration felt through steering wheel or floor at specific speeds
- High-frequency vibration (1000-5000 Hz) detectable through chassis
- Bearing temperature 10-15°C higher than motor body temperature
- Metallic grinding sounds during coasting (motor freewheeling)
- Torque delivery feeling uneven or jerky at constant speed

Preventive Suggestions for Drivers:

- Avoid aggressive acceleration from cold starts (allow 2-3 minutes warm-up)
- Listen for new or changing noises during quiet driving conditions
- Document vibration patterns (speed ranges, conditions when occurring)
- Avoid harsh impacts (potholes, curbs) that can damage bearings
- Inspect motor mounts during regular service for looseness
- Report any new vibration or noise immediately—early detection prevents catastrophic failure

When Service is Required:

- **STOP IMMEDIATELY**: Loud grinding/clicking, severe vibration, motor seizure feeling
- **Immediate**: Persistent bearing noise, vibration >1.0 g RMS, temp differential >15°C
- **Within 1 week**: New consistent noise, vibration 0.5-1.0 g RMS, efficiency loss >3%
- **Next scheduled service**: Minor vibration at specific speeds, slight noise increase

2.3 Rotor Eccentricity / Misalignment

Description: Uneven air gap between rotor and stator caused by bearing wear, shaft deflection, or mechanical stress, resulting in imbalanced magnetic forces and vibration.

Data Inputs Required:

- Vibration frequency analysis (harmonic signature)
- Three-phase current balance (A-B-C phases)
- Motor RPM vs. torque fluctuation
- Shaft position sensor data (if available)
- Radial force measurements
- Back-EMF waveform analysis

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Air gap variation	<5%	5-10%	>10%
Phase current imbalance	<3%	3-8%	>8%
Torque ripple frequency	<2% at 2x electrical freq	2-5%	>5%
Vibration at rotational freq	<0.3 g	0.3-0.8 g	>0.8 g

Early Warning Signs:

- Persistent vibration at specific RPM ranges (harmonic resonance)
- Phase current showing gradual imbalance over weeks
- "Clunking" sensation during acceleration/deceleration transitions
- Uneven torque delivery at low speeds
- Increased motor noise at specific speeds
- Steering wheel vibration correlated with motor speed

Preventive Suggestions for Drivers:

- Avoid harsh bumps, potholes, and off-road driving
- Inspect drivetrain mounts quarterly for wear or damage
- Report persistent vibration even if mild—misalignment worsens over time
- Avoid sudden high-torque launches that stress drivetrain
- Document speed ranges where vibration occurs
- Have wheel balance and alignment checked (can mask motor issues)

When Service is Required:

- **STOP IMMEDIATELY**: Severe vibration, loud clunking, sudden torque loss
- **Immediate**: Phase imbalance >6%, vibration >0.6 g, persistent clunking
- **Within 1 week**: Phase imbalance 4-6%, torque ripple >3%, new vibration pattern
- **Next scheduled service**: Phase imbalance 3-4%, minor vibration at specific speeds

2.4 Stator Winding Short / Insulation Breakdown

Description: Electrical insulation between stator windings degrades due to thermal stress, moisture, or contamination, causing short circuits between coils or to ground.

Data Inputs Required:

- Phase current imbalance
- Motor temperature hotspots (if multi-point sensing available)
- Insulation resistance measurements
- Leakage current to ground
- Ground fault detection signals
- Phase-to-phase resistance
- Motor efficiency trends

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Insulation resistance	>100 MΩ	10-100 MΩ	<10 MΩ
Leakage current	<1 mA	1-5 mA	>5 mA
Phase resistance imbalance	<2%	2-5%	>5%
Phase current imbalance	<5%	5-12%	>12%
Temperature hotspot	<10°C above avg	10-20°C	>20°C

Early Warning Signs:

- Gradual increase in phase current imbalance
- Localized overheating in one section of motor
- "Ground Fault" or "Insulation Fault" warnings
- Reduced motor efficiency (increased current for same torque)
- Intermittent power loss or torque reduction
- Burning smell from motor compartment

Preventive Suggestions for Drivers:

- Avoid driving through deep water or flooded roads (water ingress risk)
- Keep motor compartment clean and dry
- Address coolant leaks immediately (coolant can damage insulation)
- Park in covered areas during extended storage
- Report any "Ground Fault" warnings immediately
- Avoid high-voltage washing of underbody near motor

When Service is Required:

- **STOP IMMEDIATELY**: Ground fault warning, insulation resistance <5 MΩ, burning smell
- **Immediate**: Leakage current >3 mA, phase imbalance >10%, persistent fault codes
- **Within 1 week**: Insulation resistance 10-50 MΩ, phase imbalance 7-10%
- **Next scheduled service**: Insulation resistance 50-100 MΩ, phase imbalance 5-7%

2.5 Demagnetization of Permanent Magnets (PMSM Motors)

Description: Permanent magnets in the rotor lose magnetic strength due to excessive temperature, overcurrent, or age, reducing torque output and efficiency.

Data Inputs Required:

- Motor torque constant (Nm/A) over time
- Back-EMF voltage at constant speed
- Motor current vs. torque relationship
- Peak rotor temperature history
- Efficiency metrics (mechanical power out / electrical power in)
- Torque delivery vs. commanded torque

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Torque constant degradation	<2% from new	2-5%	>5%
Back-EMF reduction	<3% from baseline	3-7%	>7%
Current increase for same torque	<5%	5-12%	>12%
Peak rotor temperature	<120°C	120-140°C	>140°C

Early Warning Signs:

- Gradual reduction in acceleration performance
- Motor drawing more current for same power output
- Efficiency decreasing over months (kWh/km increasing)
- Reduced regenerative braking torque
- "Reduced Power" mode activating more frequently
- Back-EMF voltage lower than expected at given RPM

Preventive Suggestions for Drivers:

- Avoid sustained high-temperature operation (>110°C stator temp)
- Limit prolonged high-power demands in hot weather
- Ensure cooling system is functioning properly
- Avoid overcurrent conditions (aggressive acceleration from standstill)
- Monitor efficiency trends monthly—sudden drops indicate issues
- Keep firmware updated (inverter protection algorithms)

When Service is Required:

- **Immediate**: Torque constant degradation >6%, current increase >10%, persistent power limiting
- **Within 1 week**: Torque constant degradation 4-6%, efficiency loss >5%
- **Within 1 month**: Torque constant degradation 2-4%, efficiency loss 3-5%
- **Next scheduled service**: Torque constant degradation <2%, minor efficiency reduction

2.6 Phase Imbalance or Open Circuit

Description: One or more motor phases deliver less current due to wiring faults, connector corrosion, or inverter switching failures.

Data Inputs Required:

- Real-time three-phase current (A, B, C phases)
- Phase voltage measurements
- Torque response vs. command signal
- Inverter diagnostic trouble codes (DTCs)
- Phase resistance measurements
- Motor vibration patterns

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Phase current deviation	<5% between phases	5-15%	>15%
Phase voltage deviation	<3%	3-10%	>10%
Torque delivery	>95% of commanded	85-95%	<85%
Single-phase current	Within 10% of avg	10-30% below avg	>30% below or 0 A

Early Warning Signs:

- Sudden torque reduction during acceleration
- "Drive System Fault" or "Phase Fault" warnings
- Motor running roughly or vibrating at low speeds
- One phase showing consistently lower current
- Regenerative braking reduced or inconsistent
- Intermittent loss of power

Preventive Suggestions for Drivers:

- **STOP DRIVING** if sudden torque loss occurs
- Inspect high-voltage connectors for corrosion during service
- Report any "Phase Fault" warnings immediately
- Avoid restarting vehicle repeatedly if fault persists
- Document conditions when fault occurs (speed, power level, weather)

When Service is Required:

- **STOP IMMEDIATELY**: Phase current >20% imbalanced, torque delivery <80%, open circuit detected
- **Immediate**: Phase current 15-20% imbalanced, persistent fault codes
- **Within 1 week**: Phase current 10-15% imbalanced, intermittent faults
- **Next scheduled service**: Phase current 5-10% imbalanced, no functional impact

2.7 Cooling System Degradation

Description: Motor cooling system (pump, radiator, coolant) fails to adequately remove heat, leading to thermal stress and reduced performance.

Data Inputs Required:

- Coolant inlet/outlet temperature delta (ΔT)
- Cooling pump speed and flow rate
- Coolant temperature vs. motor temperature correlation
- Fan operation logs
- Ambient temperature
- Coolant level sensor (if available)

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Coolant ΔT (in-out)	5-15°C	15-25°C	>25°C
Pump flow rate	90-100% rated	70-90%	<70%
Motor temp rise rate	<2°C/min under load	2-4°C/min	>4°C/min
Coolant temperature	40-70°C	70-85°C	>85°C

Early Warning Signs:

- Coolant temperature delta increasing over weeks
- Motor temperature rising faster than normal under load
- Cooling pump noise or vibration
- Coolant level dropping (requires frequent top-ups)
- "Cooling System Fault" warnings
- Reduced coolant flow rate

Preventive Suggestions for Drivers:

- Check coolant level monthly when engine cold
- Inspect for coolant leaks under vehicle after parking
- Listen for cooling pump operation (should be quiet)
- Monitor coolant temperature trends
- Service cooling system per manufacturer schedule (typically 5-8 years)
- Flush and replace coolant at recommended intervals
- Inspect radiator for debris blockage

When Service is Required:

- **STOP IMMEDIATELY**: Coolant leak, pump failure, motor temp >120°C
- **Immediate**: Coolant ΔT >20°C, pump flow <75%, persistent cooling warnings
- **Within 1 week**: Coolant ΔT 15-20°C, coolant level low, pump noise
- **Next scheduled service**: Coolant ΔT 12-15°C, gradual temp increase trend

SECTION 3: POWER ELECTRONICS (INVERTER) MONITORING

The inverter converts DC battery power into variable-frequency AC power for the motor. Key components include:

- **Power semiconductors**: IGBTs or MOSFETs performing high-speed switching
- **DC link capacitors**: Smoothing voltage and providing energy storage
- **Gate drivers**: Controlling power semiconductor switching
- **Current and voltage sensors**: Monitoring electrical parameters
- **Cooling system**: Managing heat from switching losses
- **Control electronics**: Processing commands and feedback

Inverter Failure Prediction Scenarios

3.1 Inverter Overheating (Thermal Overload)

Description: Inverter temperature exceeds safe limits due to high power demands, cooling system issues, or component degradation.

Data Inputs Required:
- Inverter case temperature
- IGBT/MOSFET junction temperature (if available)
- Coolant inlet/outlet temperature
- Ambient temperature
- Motor current and power output
- Cooling pump speed and fan operation
- Vehicle speed and driving pattern
- Switching frequency

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Inverter case temp	50-75°C (122-167°F)	75-90°C (167-194°F)	>90°C (>194°F)
Junction temperature	80-110°C (176-230°F)	110-135°C (230-275°F)	>135°C (>275°F)
Coolant ΔT	8-18°C	18-28°C	>28°C
Temp rise rate	<3°C/min	3-6°C/min	>6°C/min

Early Warning Signs:
- Inverter temperature rising 8-10°C over weeks for same driving
- Temperature remaining elevated (>70°C) 20+ minutes after stopping

- Cooling fan running continuously or at high speed
- "Inverter Overheat" or "Power Limited" warnings
- Reduced power delivery when inverter is hot
- High-pitched whining from inverter area

Preventive Suggestions for Drivers:
- Avoid prolonged aggressive acceleration in hot weather
- Take cooling breaks during extended high-power driving
- Ensure inverter cooling system is serviced regularly
- Reduce power demands if temperature warnings appear
- Allow inverter to cool before fast charging after spirited driving
- Monitor coolant level and condition
- Avoid blocking underbody airflow (mud, debris)

When Service is Required:
- **STOP IMMEDIATELY**: Inverter temp >95°C, burning smell, smoke, persistent power loss
- **Immediate**: Frequent temp >85°C, cooling system leak, fan failure
- **Within 1 week**: Coolant ΔT >25°C, temp rising trend, frequent warnings
- **Next scheduled service**: Occasional temp 75-85°C, gradual increase

3.2 DC Bus Overvoltage / Undervoltage

Description: DC link voltage becomes unstable due to regenerative braking spikes, battery connection issues, or capacitor degradation.

Data Inputs Required:
- DC bus voltage (real-time)
- Battery pack voltage and SOC
- Regenerative braking current
- Braking intensity and frequency
- DC link capacitor voltage ripple
- Diagnostic trouble codes for voltage faults

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
DC bus voltage	300-420V (model-specific)	280-300V or 420-450V	<280V or >450V
Voltage ripple	<5% of nominal	5-10%	>10%
Regen voltage spike	<30V above nominal	30-50V	>50V
Voltage sag under load	<20V below nominal	20-40V	>40V
Voltage deviation rate	<50V/sec	50-100V/sec	>100V/sec

Early Warning Signs:
- Voltage spikes during regenerative braking (>430V)

- Voltage sags during hard acceleration (<290V)
- "Overvoltage" or "Undervoltage" fault codes
- Regenerative braking cutting out prematurely
- Power delivery inconsistent or jerky
- DC voltage fluctuating more than ±20V during steady cruise

Preventive Suggestions for Drivers:
- Avoid abrupt, maximum regenerative braking (especially downhill)
- Reduce regen intensity in cold weather when battery can't accept full power
- Check battery connection terminals during service
- Keep vehicle software updated (power management algorithms)
- Use moderate braking rather than maximum regen
- Monitor for correlation between braking and voltage warnings
- Avoid charging to 100% before long downhill drives (limits regen capacity)

When Service is Required:
- **STOP IMMEDIATELY**: DC voltage >460V or <270V, persistent fault, vehicle won't drive
- **Immediate**: Frequent overvoltage events, voltage spikes >440V, regen disabled
- **Within 1 week**: Voltage ripple >8%, occasional overvoltage warnings
- **Next scheduled service**: Voltage ripple 5-8%, minor voltage fluctuations

3.3 Phase Current Imbalance

Description: Three-phase output currents become unbalanced due to gate driver faults, IGBT/MOSFET degradation, or motor winding issues.

Data Inputs Required:
- Three-phase current measurements (A, B, C)
- Motor torque and speed feedback
- Inverter PWM signal statistics
- Total harmonic distortion (THD)
- Phase voltage measurements
- Switching frequency stability

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Phase current deviation	<3% between phases	3-8%	>8%
THD (Total Harmonic Distortion)	<5%	5-10%	>10%
Torque ripple	<3%	3-7%	>7%
Switching frequency variation	±5%	±5-10%	>±10%

Early Warning Signs:
- Gradual increase in phase current imbalance over weeks

- Motor vibration or rough operation at low speeds
- "Torque Deviation" or "Phase Fault" warnings
- Uneven torque delivery during acceleration
- One phase consistently carrying more/less current
- Increased motor noise or whining

Preventive Suggestions for Drivers:
- Report any new vibration or rough motor operation
- Avoid heavy torque demands if phase imbalance detected
- Document conditions when torque irregularities occur
- Schedule service if "Phase Fault" warnings appear
- Monitor torque delivery consistency during acceleration tests

When Service is Required:
- **STOP IMMEDIATELY**: Phase imbalance >12%, severe torque ripple, motor won't operate smoothly
- **Immediate**: Phase imbalance 8-12%, persistent torque deviation warnings
- **Within 1 week**: Phase imbalance 5-8%, THD >8%, noticeable torque irregularity
- **Next scheduled service**: Phase imbalance 3-5%, minor torque ripple

3.4 IGBT/MOSFET Degradation (Switching Failure)

Description: Power semiconductors degrade due to thermal cycling, electrical overstress, or age, leading to increased switching losses and potential failure.

Data Inputs Required:
- Junction temperature history
- Thermal cycles per trip
- Switching frequency and gate voltage
- Current waveform distortion
- Inverter efficiency (power in vs. power out)
- On-state voltage drop (Vce or Vds)

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Inverter efficiency	>95%	90-95%	<90%
On-state voltage drop	<2V	2-3V	>3V
Switching loss increase	<5% from baseline	5-15%	>15%
Junction temp rise	<10°C for same load	10-20°C	>20°C
Thermal cycles per day	<50	50-100	>100

Early Warning Signs:
- Inverter efficiency decreasing over months

- More heat generation for same power output
- Junction temperature rising for same operating conditions
- Current waveform showing increased distortion
- "Inverter Degradation" or efficiency warnings
- Increased switching noise

Preventive Suggestions for Drivers:
- Limit repeated full-power accelerations (reduces thermal stress)
- Avoid aggressive driving in extreme temperatures
- Allow gradual warm-up before high-power demands
- Monitor efficiency trends monthly (kWh/km or mi/kWh)
- Keep inverter firmware updated (optimizes switching control)
- Service cooling system to minimize thermal cycling

When Service is Required:
- **STOP IMMEDIATELY**: Efficiency <85%, inverter shutdown, burning smell
- **Immediate**: Efficiency 85-90%, junction temp >130°C, persistent degradation warnings
- **Within 1 week**: Efficiency 90-93%, on-state voltage >2.5V, temp rise >15°C
- **Next scheduled service**: Efficiency 93-95%, minor efficiency reduction trend

3.5 DC Link Capacitor Aging

Description: Electrolytic or film capacitors lose capacitance over time, causing increased voltage ripple and reduced filtering capability.

Data Inputs Required:
- DC bus voltage ripple amplitude
- Capacitor equivalent series resistance (ESR)
- Temperature cycling history
- Total inverter operating hours
- Ripple current measurements
- Capacitor case temperature

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Voltage ripple	<5% of DC voltage	5-10%	>10%
Capacitance loss	<10% from new	10-20%	>20%
ESR increase	<50% from new	50-100%	>100%
Ripple current	<Rated value	Rated to 120%	>120% rated

Early Warning Signs:
- DC bus voltage ripple increasing gradually
- Capacitor temperature rising (>10°C above ambient)

- Increased inverter noise or humming
- Voltage instability during power transitions
- "DC Link Fault" warnings
- Reduced power quality

Preventive Suggestions for Drivers:
- Avoid sustained high-power operation in hot weather (accelerates aging)
- Monitor inverter operating hours (capacitors have service life)
- Schedule capacitor inspection/replacement per OEM schedule (typically 100K miles or 8-10 years)
- Report increased electrical noise from inverter area
- Avoid extreme temperature exposure when possible

When Service is Required:
- **STOP IMMEDIATELY**: Voltage ripple >15%, capacitor bulging/leaking, inverter shutdown
- **Immediate**: Voltage ripple 10-15%, ESR >80% increase, persistent DC link faults
- **Within 1 month**: Voltage ripple 7-10%, capacitance loss 15-20%, approaching service life
- **Next scheduled service**: Voltage ripple 5-7%, capacitance loss 10-15%

3.6 Insulation Breakdown / Ground Fault

Description: Electrical insulation degrades due to moisture, contamination, or thermal stress, causing leakage currents to chassis ground.

Data Inputs Required:
- Leakage current to ground
- Insulation resistance measurements
- Ambient humidity and temperature
- Ground fault detection signals
- High-voltage interlock status
- Diagnostic trouble codes

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Insulation resistance	>500 kΩ/V	100-500 kΩ/V	<100 kΩ/V
Leakage current	<0.5 mA	0.5-2 mA	>2 mA
Ground fault threshold	0 faults/month	1-3 faults/month	>3 faults/month

Early Warning Signs:
- "Ground Fault" or "Insulation Fault" warnings
- Intermittent high-voltage system shutdown
- Leakage current increasing over time
- Insulation resistance decreasing gradually
- Faults occurring after wet weather or car wash

- Tingling sensation when touching vehicle (rare but serious)

Preventive Suggestions for Drivers:
- **DO NOT DRIVE** if ground fault warning persists
- Avoid driving through deep water or flooded areas
- Avoid high-pressure washing of underbody near inverter
- Park in dry, covered areas during extended storage
- Report any "Ground Fault" warnings immediately
- Inspect inverter housing for moisture or damage

When Service is Required:
- **STOP IMMEDIATELY**: Ground fault detected, insulation resistance <50 kΩ/V, leakage >5 mA, tingling sensation
- **Immediate**: Persistent ground fault warnings, insulation resistance 50-100 kΩ/V
- **Within 1 week**: Intermittent ground faults, insulation resistance 100-200 kΩ/V
- **Next scheduled service**: Insulation resistance 200-500 kΩ/V, single fault occurrence

3.7 Connector Corrosion or Contact Resistance

Description: High-voltage connectors develop corrosion or increased contact resistance due to environmental exposure, causing localized heating and power loss.

Data Inputs Required:
- Connector temperature sensors
- Voltage drop across connectors
- Current flow measurements
- Environmental humidity and salt exposure data (GPS + weather API)
- Contact resistance measurements
- Power delivery efficiency

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Connector temperature	<50°C	50-70°C	>70°C
Voltage drop	<0.5V at rated current	0.5-1.5V	>1.5V
Contact resistance	<1 mΩ	1-5 mΩ	>5 mΩ
Temp rise above ambient	<15°C	15-30°C	>30°C

Early Warning Signs:
- Connector temperature rising over weeks
- Localized heating detected during inspection
- Voltage drop increasing across connections
- Power delivery reduced at high current
- Visible corrosion or discoloration on connectors

- "Connection Fault" warnings

Preventive Suggestions for Drivers:
- Avoid parking in high-humidity or coastal environments when possible
- Inspect high-voltage connectors annually for corrosion
- Keep underbody clean from road salt (especially winter)
- Report any unusual heating near connector areas
- Avoid water crossings that submerge connectors
- Apply dielectric grease during service (technician task)

When Service is Required:
- **STOP IMMEDIATELY**: Connector temp >80°C, burning smell, visible arcing
- **Immediate**: Connector temp 70-80°C, voltage drop >1.2V, visible corrosion
- **Within 1 week**: Connector temp 60-70°C, voltage drop 0.8-1.2V
- **Next scheduled service**: Connector temp 50-60°C, voltage drop 0.5-0.8V

SECTION 4: ON-BOARD CHARGER (OBC) MONITORING

The On-Board Charger converts AC power from charging stations into DC power for the battery. Components include:

- **AC input stage**: Power factor correction and rectification
- **DC-DC converter**: Voltage regulation and current control
- **Communication module**: Negotiates with EVSE (charging station)
- **Safety circuits**: Ground fault and overcurrent protection
- **Cooling system**: Manages heat during charging

OBC Failure Prediction Scenarios

4.1 Charging Rate Degradation

Description: OBC unable to deliver rated charging power due to component aging, thermal issues, or control circuit problems.

Data Inputs Required:
- Actual charging power vs. rated power
- Charging current and voltage
- OBC temperature during charging
- AC input voltage and frequency
- Power factor measurements

- Charging session duration trends
- EVSE communication logs

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Charging power delivery	90-100% of rated	75-90%	<75%
Charging efficiency	>92%	88-92%	<88%
Charge time deviation	±5% from expected	±5-15%	>±15%
Power factor	>0.95	0.90-0.95	<0.90

Early Warning Signs:
- Charging sessions taking 10-15% longer than normal
- Charging power 10-20% below EVSE rating
- Frequent charging session interruptions
- "Charging Fault" or "Reduced Charging Power" warnings
- OBC temperature higher than normal during charging
- Inability to negotiate maximum power with EVSE

Preventive Suggestions for Drivers:
- Document charging times for consistent sessions (e.g., 20% → 80%)
- Monitor actual charging power vs. EVSE rating
- Ensure charging connectors are clean and dry
- Verify EVSE is functioning properly (test with another vehicle if possible)
- Keep OBC firmware updated
- Report gradual charging time increases

When Service is Required:
- **Immediate**: Charging power <70% of rated, frequent session failures, OBC temp >85°C
- **Within 1 week**: Charging power 70-80%, charge time increased >20%, efficiency <90%
- **Within 1 month**: Charging power 80-90%, charge time increased 10-20%
- **Next scheduled service**: Charging power 90-95%, minor efficiency reduction

4.2 OBC Overheating

Description: Excessive heat generation during charging due to component degradation, cooling system issues, or high ambient temperature.

Data Inputs Required:
- OBC case temperature
- Power semiconductor junction temperature
- Coolant temperature (if liquid-cooled)
- Ambient temperature
- Charging power level
- Cooling fan speed and operation

- Thermal derating events

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
OBC temperature	40-65°C (104-149°F)	65-80°C (149-176°F)	>80°C (>176°F)
Temp rise above ambient	<30°C	30-45°C	>45°C
Thermal derating events	0 per month	1-3 per month	>3 per month
Cooling fan duty cycle	<60%	60-85%	>85%

Early Warning Signs:
- OBC temperature rising over weeks for same charging conditions
- Charging power reduced due to thermal limiting
- "OBC Overheat" warnings during normal charging
- Cooling fan running continuously at high speed
- Charging sessions interrupted due to temperature
- OBC remaining hot (>55°C) 30+ minutes after charging

Preventive Suggestions for Drivers:
- Avoid charging immediately after high-speed driving (allow cooling)
- Charge during cooler parts of day in hot weather
- Ensure OBC cooling vents are not blocked
- Park in shade during daytime charging when possible
- Reduce charging power if overheating warnings appear
- Monitor coolant level if liquid-cooled system

When Service is Required:
- **STOP CHARGING**: OBC temp >85°C, burning smell, charging repeatedly interrupted
- **Immediate**: Frequent thermal derating, OBC temp consistently >75°C, cooling fan failure
- **Within 1 week**: OBC temp 70-75°C during normal charging, multiple derating events
- **Next scheduled service**: OBC temp 65-70°C, occasional thermal limiting

4.3 Power Factor and Efficiency Degradation

Description: OBC power factor decreases due to component aging or control issues, increasing reactive power and reducing efficiency.

Data Inputs Required:
- Input power (kW) from EVSE
- Output power (kW) to battery
- Power factor measurement
- Input current harmonics (THD - Total Harmonic Distortion)
- Reactive power consumption
- OBC temperature

- Charging efficiency trend over time

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Power factor	**>0.95**	**0.90-0.95**	**<0.90**
Charging efficiency	**>92%**	**88-92%**	**<88%**
Input current THD	**<5%**	**5-10%**	**>10%**
Reactive power	**<10% of real power**	**10-20%**	**>20%**
Energy loss as heat	**<8%**	**8-12%**	**>12%**

Early Warning Signs:
- Power factor decreasing from >0.95 to 0.90-0.93 over months
- Charging efficiency dropping (more kWh from wall than stored in battery)
- OBC generating more heat for same charging power
- EVSE circuit breaker tripping more frequently
- Input current waveform distortion visible on monitoring equipment
- Utility billing showing higher energy consumption for same charging

Preventive Suggestions for Drivers:
- Monitor "energy added" vs. "energy billed" using smart meters
- Calculate charging efficiency: (Battery kWh added / Wall kWh consumed) × 100%
- Document efficiency monthly to identify degradation trends
- Report efficiency drops >3% from baseline
- Ensure electrical service is adequate (proper voltage, grounding)
- Avoid charging on circuits with poor power quality
- Have licensed electrician verify home charging circuit

When Service is Required:
- Immediate: Power factor <0.85, efficiency <85%, repeated breaker trips
- Within 1 week: Power factor 0.85-0.90, efficiency 85-88%, THD >8%
- Within 1 month: Power factor 0.90-0.93, efficiency 88-90%
- Next scheduled service: Power factor 0.93-0.95, efficiency 90-92%

4.4 Communication and Control Faults

Description: OBC fails to properly communicate with EVSE or battery management system, causing charging failures or safety issues.

Data Inputs Required:
- EVSE communication error logs

- BMS communication status
- Charging session start/stop success rate
- Pilot signal quality (SAE J1772 or IEC 61851)
- Diagnostic trouble codes
- Charging session interruption frequency
- Software/firmware version

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Communication errors	0 per 100 sessions	1-5 per 100 sessions	>5 per 100 sessions
Session start failures	<1%	1-5%	>5%
Mid-session interruptions	<2%	2-8%	>8%
Pilot signal quality	±1V of nominal	±1-2V	>±2V

Early Warning Signs:
- Charging sessions failing to start (plug in, no response)
- "Communication Error" messages on vehicle or EVSE
- Charging stopping and restarting multiple times per session
- Different EVSE stations showing same communication issues
- Intermittent "Charger Fault" warnings
- Charging port indicator light not illuminating properly

Preventive Suggestions for Drivers:
- Keep vehicle software updated (OTA or dealer service)
- Clean charging port contacts with dry cloth monthly
- Inspect charging cable for damage or corrosion
- Try multiple EVSE stations to isolate vehicle vs. station issues
- Document error messages with photos and timestamps
- Ensure charging cable is fully inserted and locked
- Avoid forcing or wiggling cable during charging
- Report intermittent communication errors even if charging eventually works

When Service is Required:
- **Immediate**: Charging impossible, persistent communication failures, safety interlocks failing
- **Within 1 week**: >10% session failures, frequent interruptions, pilot signal errors
- **Within 1 month**: 5-10% session failures, occasional interruptions
- **Next scheduled service**: <5% session failures, minor communication delays

4.5 Input Voltage Instability

Description: OBC struggles with inconsistent input voltage from electrical service, causing charging interruptions or damage.

Data Inputs Required:

- Input AC voltage (real-time)
- Voltage fluctuation magnitude and frequency
- Input current draw
- Charging power variations
- Session interruption correlation with voltage events
- Electrical service quality metrics

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Input voltage (120V)	114-126V	108-114V or 126-132V	<108V or >132V
Input voltage (240V)	228-252V	216-228V or 252-264V	<216V or >264V
Voltage fluctuation	<3%	3-6%	>6%
Voltage sag under load	<5V	5-10V	>10V

Early Warning Signs:

- Charging power fluctuating during session
- Charging interruptions during high electrical demand periods
- OBC shutting down with "Input Voltage Fault"
- Lights dimming when charging starts
- Inconsistent charging times for same SOC range
- Circuit breaker trips when other appliances operate

Preventive Suggestions for Drivers:

- Have licensed electrician verify home electrical service capacity
- Ensure dedicated circuit for EV charging (40A minimum for Level 2)
- Avoid charging during peak electrical demand if voltage issues occur
- Install voltage monitoring equipment to document issues
- Consider voltage stabilizer if utility service is poor
- Report voltage issues to utility company
- Use lower charging rate if voltage problems persist

When Service is Required:

- **Immediate**: Persistent voltage faults, charging impossible, electrical hazards
- **Within 1 week**: Frequent voltage-related interruptions, voltage >260V or <210V
- **Within 1 month**: Occasional voltage issues, voltage 216-220V or 252-260V
- **Electrical service issue**: If OBC tests normal but voltage problems persist, address electrical service

SECTION 5: TIRE SYSTEM MONITORING

Tire health is critical for EV safety, efficiency, and range. Monitoring includes:

- **Tire Pressure Monitoring System (TPMS)**: Sensors measuring pressure and temperature

- **Tread depth**: Remaining rubber depth for traction
- **Wear patterns**: Indicating alignment, inflation, or suspension issues
- **Temperature**: Heat generation from friction and load
- **Structural integrity**: Sidewall condition, bulges, damage

Tire Failure Prediction Scenarios

5.1 Rapid Air Pressure Loss (Puncture/Blowout)

Description: Sudden and sharp drop in tire pressure due to puncture, valve failure, or catastrophic tire failure.

Data Inputs Required:
- Real-time TPMS pressure data (all four tires)
- Pressure rate of change (ΔP/second)
- Tire temperature
- Wheel speed sensor data (detect imbalance)
- Vehicle speed and yaw rate
- Impact detection (accelerometer spikes)

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Pressure drop rate	<0.5 PSI/min	0.5-2 PSI/min	>2 PSI/min
Absolute pressure	32-36 PSI (typical)	28-32 PSI	<28 PSI
Pressure deviation	<2 PSI between tires	2-4 PSI	>4 PSI
Wheel speed deviation	<2%	2-5%	>5%

Early Warning Signs:
- TPMS warning light illuminated
- Rapid pressure drop visible on display (>1 PSI/minute)
- Vehicle pulling to one side
- Steering feels unstable or heavy
- Unusual vibration or thumping sound
- Wheel speed sensor showing one tire rotating differently

Preventive Suggestions for Drivers:
- **STOP IMMEDIATELY** if rapid pressure loss detected
- Check tire pressure before every trip (visual inspection minimum)
- Inspect tires for embedded objects weekly
- Avoid driving over debris, potholes, or sharp objects
- Replace valve caps to prevent valve damage
- Carry tire repair kit and portable air compressor
- Know location of spare tire or run-flat capability

When Service is Required:

- **STOP IMMEDIATELY**: Pressure drop >2 PSI/min, pressure <25 PSI, steering instability
- **Pull over safely**: Pressure 25-28 PSI, visible damage, persistent TPMS warning
- **Service within 24 hours**: Slow leak requiring daily inflation, pressure 28-30 PSI

5.2 Slow Pressure Leak

Description: Gradual pressure loss over hours/days from small punctures, valve leaks, or rim seal issues.

Data Inputs Required:

- TPMS pressure trend over time (hours/days)
- Ambient temperature (to normalize readings)
- Maintenance logs (refill history)
- Pressure loss rate (PSI/day)
- Temperature-corrected pressure

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Pressure loss rate	<0.5 PSI/week	0.5-2 PSI/week	>2 PSI/week
Refill frequency	<1x per month	1-2x per month	>2x per month
Pressure after 24 hours	>95% of filled	90-95%	<90%

Early Warning Signs:

- One tire consistently lower pressure than others
- Requiring weekly pressure adjustments
- Pressure dropping 1-3 PSI over 2-3 days
- TPMS warning appearing intermittently
- Tire pressure lower in morning (cold) than expected

Preventive Suggestions for Drivers:

- Check tire pressure weekly (all tires, including spare)
- Document pressure readings to identify trends
- Inspect tire for embedded objects (nails, screws)
- Check valve stem for leaks (apply soapy water, look for bubbles)
- Address slow leaks promptly—they worsen over time
- Have tire professionally inspected if leak source unclear
- Rotate tires per schedule to ensure even wear

When Service is Required:

- **Within 1 week**: Pressure loss >2 PSI/week, requiring 2+ refills/week
- **Within 2 weeks**: Pressure loss 1-2 PSI/week, visible embedded object
- **Next scheduled service**: Pressure loss 0.5-1 PSI/week, no visible cause

5.3 Chronic Under-Inflation

Description: Tire consistently operates below recommended pressure, increasing rolling resistance, heat generation, and wear.

Data Inputs Required:

- TPMS pressure data
- Tire temperature (increase

d temp under load)

- Fuel/energy consumption metrics
- Tire pressure vs. manufacturer specification
- Duration at low pressure
- Load conditions (vehicle weight, cargo)

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Pressure vs. recommended	95-105%	85-95%	<85%
Absolute pressure	32-36 PSI (typical)	28-32 PSI	<28 PSI
Temperature rise	<15°C above ambient	15-25°C	>25°C
Energy consumption increase	<2%	2-5%	>5%
Time at low pressure	0 hours	<24 hours	>24 hours

Early Warning Signs:

- TPMS warning light illuminated consistently
- One or more tires 3-5 PSI below specification
- Increased energy consumption (kWh/km or kWh/mile)
- Reduced vehicle range for same driving conditions
- Uneven tire wear (more wear on outer edges)
- Tire sidewalls appear bulged or flexed excessively
- Handling feels soft or imprecise

Preventive Suggestions for Drivers:

- Check tire pressure monthly minimum (weekly recommended)
- Inflate to manufacturer specification (door placard or manual)
- Check pressure when tires are cold (before driving)
- Adjust pressure for load (heavy cargo requires higher pressure)
- Use quality pressure gauge (digital recommended)
- Don't rely solely on TPMS—manual checks essential
- Understand that cold weather drops pressure (~1 PSI per 10°F drop)
- Set calendar reminders for pressure checks

When Service is Required:

- **Immediate**: Pressure <85% spec for >24 hours, visible tire damage
- **Within 1 week**: Pressure 85-90% spec, uneven wear developing
- **Within 2 weeks**: Pressure 90-95% spec, energy consumption increased >3%

- **Next scheduled service**: Pressure 95-98% spec, minor efficiency impact

5.4 Over-Inflation

Description: Excessive air pressure reduces tire contact patch, causing harsh ride, reduced traction, and uneven center wear.

Data Inputs Required:
- TPMS pressure readings
- Suspension vibration/vertical acceleration
- Tire contact patch pressure distribution (if available)
- Tire wear pattern data
- Ambient temperature

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Pressure vs. max sidewall	<90% of max	90-95% of max	>95% of max
Pressure vs. recommended	95-105%	105-115%	>115%
Ride harshness index	Baseline	+10-25%	>+25%
Center tread wear rate	Even with edges	10-20% faster	>20% faster

Early Warning Signs:
- Tire pressure 3-6 PSI above manufacturer specification
- Harsh, jarring ride quality
- Reduced traction in wet conditions
- Tire center wearing faster than edges
- Suspension vibration increased
- Tire appearing overly rigid or "balloon-like"

Preventive Suggestions for Drivers:
- Don't exceed manufacturer recommended pressure
- Don't inflate to maximum sidewall pressure (that's max, not recommended)
- Adjust pressure when ambient temperature increases significantly
- Check pressure after temperature changes (hot day can increase 2-4 PSI)
- Release air if pressure exceeds specification
- Avoid "topping off" tires without checking actual pressure
- Understand recommended pressure is for cold tires

When Service is Required:
- **Within 1 week**: Pressure >110% spec, center wear developing
- **Within 2 weeks**: Pressure 105-110% spec, ride quality significantly affected
- **Next scheduled service**: Pressure 100-105% spec, minor wear pattern changes

5.5 Excessive Heat Build-Up

Description: Tire overheating due to prolonged high-speed driving, overload, under-inflation, or brake drag.

Data Inputs Required:

- Tire temperature sensors (TPMS with temp capability)
- Ambient temperature
- Vehicle speed and duration
- Vehicle load (weight sensors or suspension data)
- Brake usage patterns
- Tire pressure correlation

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Tire temperature	Ambient +10-30°C	Ambient +30-50°C	>Ambient +50°C
Absolute temperature	<70°C (158°F)	70-90°C (158-194°F)	>90°C (>194°F)
Temp rise rate	<2°C/10 min	2-4°C/10 min	>4°C/10 min
Temp differential (L-R)	<10°C	10-20°C	>20°C

Early Warning Signs:

- TPMS showing elevated tire temperatures
- One tire significantly hotter than others (brake drag)
- Tire temperature >80°C during normal driving
- Burning rubber smell
- Tire pressure increasing >4 PSI during driving
- Tire remaining hot (>50°C) 30+ minutes after stopping

Preventive Suggestions for Drivers:

- Take 10-15 minute cooling breaks during extended high-speed driving
- Avoid sustained speeds >120 km/h (75 mph) in hot weather (>35°C)
- Check tire pressure before long trips—under-inflation causes overheating
- Reduce speed if tire temperature warnings appear
- Don't overload vehicle beyond GVWR (Gross Vehicle Weight Rating)
- Inspect brakes if one tire consistently hotter (possible drag)
- Allow tires to cool before checking/adjusting pressure

When Service is Required:

- **STOP IMMEDIATELY**: Tire temp >95°C, burning smell, smoke, rapid pressure increase
- **Immediate**: Persistent temp >85°C, one tire >20°C hotter than others
- **Within 1 week**: Frequent temp warnings, temp consistently 70-85°C
- **Next scheduled service**: Occasional elevated temps, gradual increase trend

5.6 Uneven Tire Wear (Alignment/Suspension Issues)

Description: Asymmetric wear patterns indicating wheel alignment problems, suspension damage, or improper inflation.

Data Inputs Required:
- Wheel alignment data (camber, caster, toe)
- Suspension health sensors (ride height, damper performance)
- TPMS pressure and temperature (slightly higher temp on misaligned tire)
- Tire rotation history
- Tread depth measurements (if automated sensors available)
- Steering angle sensor data

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Tread depth difference	<1mm across tire	1-2mm	>2mm
Alignment toe angle	±0.05°	±0.05-0.15°	>±0.15°
Camber angle	±0.5°	±0.5-1.5°	>±1.5°
Wear rate difference	<10% L vs R	10-25%	>25%

Early Warning Signs:
- Inner or outer edge wearing faster than center
- Feathering pattern on tread blocks
- Vehicle pulling to one side
- Steering wheel off-center when driving straight
- Uneven tire temperatures (misaligned tire runs hotter)
- Vibration at specific speeds
- Tire noise increasing (roaring or humming)

Preventive Suggestions for Drivers:
- Rotate tires every 8,000-10,000 km (5,000-6,000 miles)
- Check wheel alignment every 12 months or after impact
- Inspect tires monthly for uneven wear patterns
- Address pulling or vibration immediately
- Maintain proper tire pressure (under-inflation causes edge wear)
- Avoid hitting curbs or potholes at speed
- Have suspension inspected if wear patterns develop
- Document tread depth at each rotation to track wear rates

When Service is Required:
- **Immediate**: Severe pulling, tread depth difference >2mm, safety concern
- **Within 1 week**: Visible uneven wear, tread depth difference 1-2mm
- **Within 1 month**: Minor wear pattern developing, alignment suspected

- **Next scheduled service**: Slight wear differences, preventive alignment check

5.7 Sidewall Damage (Curb/Pothole Impact)

Description: Structural damage to tire sidewall from hitting curbs, potholes, or road debris, weakening tire integrity.

Data Inputs Required:
- Accelerometer data (impact detection)
- Wheel speed fluctuations post-impact
- TPMS pressure drop after impact event
- GPS location (road quality correlation)
- Suspension compression data
- Impact severity index

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Impact force	<2g vertical	2-4g	>4g
Pressure drop post-impact	0 PSI	0-2 PSI	>2 PSI
Wheel speed anomaly	<1% variation	1-3%	>3%
Visible damage	None	Minor scuff/bulge	Bulge/cut/separation

Early Warning Signs:
- Bulge or bubble on tire sidewall
- Cuts, gouges, or exposed cords
- Vibration after hitting pothole or curb
- Pressure loss after impact
- Visible sidewall deformation
- Tire "thumping" sound while driving
- Vehicle pulling after impact

Preventive Suggestions for Drivers:
- **INSPECT IMMEDIATELY** after significant impact
- Avoid high-speed curb contact when parking
- Slow down for potholes—don't swerve dangerously
- Drive cautiously on rough or damaged roads
- Inspect sidewalls weekly for bulges or damage
- Replace tire immediately if structural damage visible
- Don't attempt to repair sidewall damage (tire must be replaced)
- Report road hazards to local authorities

When Service is Required:
- **STOP IMMEDIATELY**: Visible bulge, cut exposing cords, rapid pressure loss
- **Do not drive**: Any sidewall structural damage, separation visible

- **Immediate inspection**: Pressure drop after impact, vibration, pulling
- **Within 24 hours**: Minor sidewall scuff after impact (verify no internal damage)

5.8 Tire Imbalance or Structural Deformation

Description: Internal belt separation, bulges, or wheel imbalance causing vibration and irregular rotation.

Data Inputs Required:
- Wheel speed sensors (detect rotation irregularities)
- Vibration/accelerometer data (frequency analysis)
- Tire temperature (localized hot spots indicate separation)
- Steering wheel vibration sensors
- Speed correlation with vibration

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Wheel balance	<10g imbalance	10-30g	>30g
Vibration amplitude	<0.2g at speed	0.2-0.5g	>0.5g
Temp hot spot	<5°C above avg	5-15°C	>15°C
Runout (wobble)	<1mm	1-3mm	>3mm

Early Warning Signs:
- Vibration felt through steering wheel at 80-100 km/h (50-60 mph)
- Vibration through seat/floor at specific speeds
- Steering wheel shaking during braking
- Uneven tire wear pattern (cupping or scalloping)
- Localized heating on tire surface
- Visible wobble when tire rotates
- Thumping sound at low speeds

Preventive Suggestions for Drivers:
- Balance tires every 8,000-10,000 km (5,000-6,000 miles)
- Balance after tire installation or rotation
- Inspect for bulges or deformations monthly
- Address vibration immediately—imbalance worsens over time
- Have wheels checked for bends after impact
- Replace tires showing belt separation or bulges
- Don't drive with severe vibration (can damage suspension)

When Service is Required:
- **Immediate**: Visible bulge/deformation, severe vibration, hot spot >15°C
- **Within 1 week**: Persistent vibration, imbalance suspected, localized heating
- **Within 2 weeks**: Minor vibration at specific speeds, recent impact

- **Next scheduled service**: Slight vibration, routine balance needed

5.9 Tread Separation or Delamination

Description: Tread layer separating from tire carcass, leading to catastrophic failure, typically from manufacturing defect or age.

Data Inputs Required:
- Tire temperature sensors (localized overheating at separation point)
- Vibration data (frequency spikes at high speed)
- TPMS anomalies (pressure fluctuations)
- Tire age (manufacture date/DOT code)
- Mileage and usage patterns

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Tread adhesion	100% bonded	Partial separation	Visible separation
Temp at separation	N/A	+10-20°C localized	>+20°C
Vibration frequency	Baseline	+20-50% amplitude	>+50%
Tire age	<6 years	6-8 years	>8 years

Early Warning Signs:
- Visible tread lifting or separation from sidewall
- Bulge or bubble under tread surface
- Vibration increasing with speed
- Thumping or flapping sound
- Localized overheating on tire surface
- Tread blocks appearing loose or moving
- Cracks between tread and sidewall

Preventive Suggestions for Drivers:
- **STOP IMMEDIATELY** if tread separation suspected
- Inspect tread depth and adhesion regularly
- Replace tires older than 6 years regardless of tread depth
- Check tire manufacture date (DOT code on sidewall)
- Avoid high-speed driving on aging or worn tires
- Replace tires showing any separation or delamination
- Don't use retreaded tires on EVs (high torque stress)
- Store spare tire properly (cool, dry, away from sunlight)

When Service is Required:
- **STOP IMMEDIATELY**: Visible separation, flapping sound, severe vibration
- **Do not drive**: Any tread separation visible, bulge under tread
- **Replace immediately**: Tires >8 years old, cracks between tread and body

- **Within 1 week**: Tires 6-8 years old, minor separation signs

5.10 Valve Stem or Rim Leak

Description: Slow pressure loss from valve stem damage, corrosion, or rim seal failure.

Data Inputs Required:
- TPMS slow pressure loss detection
- Pressure loss rate and pattern
- Environmental humidity/temperature
- Valve stem age and condition
- Rim condition data

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Pressure loss rate	<0.3 PSI/week	0.3-1 PSI/week	>1 PSI/week
Valve stem age	<5 years	5-8 years	>8 years
Rim seal integrity	100% sealed	Minor leak	Significant leak

Early Warning Signs:
- Consistent slow pressure loss from one tire
- Hissing sound near valve stem
- Soapy water test shows bubbles at valve or rim
- Valve stem appears cracked or damaged
- Corrosion visible on valve or rim
- Pressure loss accelerates in cold weather

Preventive Suggestions for Drivers:
- Replace valve stems during tire changes
- Keep valve caps installed (prevents dirt/moisture)
- Inspect valve stems for cracks or damage
- Clean rim bead area before mounting tires
- Avoid hitting curbs (can damage rim seal)
- Test valve stems with soapy water if slow leak suspected
- Replace rubber valve stems every 5 years
- Use metal valve stems for high-performance applications

When Service is Required:
- **Within 1 week**: Pressure loss >1 PSI/week, visible valve damage, hissing sound
- **Within 2 weeks**: Pressure loss 0.5-1 PSI/week, valve stem >8 years old
- **Next tire service**: Pressure loss 0.3-0.5 PSI/week, preventive valve replacement
- **During tire change**: Always replace valve stems when mounting new tires

5.11 Tire Aging (Rubber Degradation/Cracking)

Description: Rubber deterioration due to UV exposure, ozone, or age, weakening tire structure even with adequate tread.

Data Inputs Required:

- Tire age (manufacture date from DOT code)
- Usage patterns (mileage, temperature cycles)
- TPMS stability (older tires may show inconsistent readings)
- Storage conditions (indoor vs. outdoor)
- UV exposure history (GPS location data)
- Climate data (temperature, humidity, ozone levels)

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Tire age	<6 years	6-8 years	>8 years
Visible cracking	None	Minor surface cracks	Deep cracks/dry rot
Rubber hardness	60-70 Shore A	70-80 Shore A	>80 Shore A
Sidewall flexibility	Normal flex	Reduced flex	Stiff/brittle

Early Warning Signs:

- Fine cracks in sidewall or tread grooves
- Tire rubber feels hard or brittle
- Whitish or grayish appearance (weathering)
- Reduced grip in wet conditions
- TPMS readings becoming less stable
- Sidewall showing "dry rot" or crazing
- Tread separating from sidewall at base

Preventive Suggestions for Drivers:

- Replace tires older than 6 years regardless of tread depth
- Check manufacture date: DOT code last 4 digits (WWYY format)
    - Example: DOT XXXX XXXX 2319 = Week 23 of 2019
- Store vehicle away from direct sunlight and ozone sources
- Use tire covers if storing vehicle outdoors long-term
- Inspect for cracks every 3 months on tires >5 years old
- Don't use "tire shine" products excessively (can accelerate aging)
- Maintain proper inflation (under-inflation accelerates aging)
- Park in garage or covered area when possible

When Service is Required:

- **Replace immediately**: Tires >10 years old, deep cracks, visible dry rot
- **Within 1 month**: Tires 8-10 years old, moderate cracking developing
- **Within 3 months**: Tires 6-8 years old, minor surface cracks
- **Plan replacement**: Tires 5-6 years old, no visible damage (proactive)

5.12 Overload or High Payload Operation

Description: Excess weight increases tire stress, pressure, and temperature, accelerating wear or causing failure.

Data Inputs Required:

- Axle load sensors or suspension compression data
- Tire temperature and pressure
- Vehicle speed and gradient (hill climbing stress)
- Load duration and frequency
- GVWR (Gross Vehicle Weight Rating) compliance
- Tire load rating vs. actual load

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Load vs. tire rating	<90%	90-100%	>100%
Load vs. GVWR	<95%	95-100%	>100%
Tire pressure increase	+2-4 PSI under load	+4-6 PSI	>+6 PSI
Tire temperature	+10-20°C under load	+20-35°C	>+35°C

Early Warning Signs:

- Tire pressure increasing significantly when loaded
- Tire temperature >80°C during loaded operation
- Sidewalls appearing excessively bulged
- Accelerated tread wear when carrying heavy loads
- Vehicle handling feeling sluggish or unstable
- Suspension bottoming out frequently
- TPMS warnings appearing when loaded

Preventive Suggestions for Drivers:

- Know your vehicle's GVWR and payload capacity
- Weigh vehicle when heavily loaded (truck scales, weigh stations)
- Increase tire pressure per load chart in owner's manual
- Distribute cargo evenly across axles
- Reduce speed when heavily loaded (especially in hot weather)
- Take more frequent breaks during loaded long-distance travel
- Inspect tires before and after heavy load operations
- Consider load-range E or higher tires for frequent heavy loads
- Don't exceed tire load rating (check sidewall)

When Service is Required:

- **Immediate**: Load >110% GVWR, tire damage from overload, handling unsafe
- **After trip**: Load 100-110% GVWR, tire temp >85°C, visible stress

- **Within 1 week**: Frequent overloading, accelerated wear visible
- **Next scheduled service**: Occasional overloading, tire inspection needed

5.13 Seasonal or Temperature-Induced Pressure Fluctuations

Description: Ambient temperature changes cause tire pressure variations, potentially leading to under-inflation in cold weather.

Data Inputs Required:
- TPMS pressure trends vs. ambient temperature
- Vehicle location (GPS-based climate data)
- Seasonal temperature variations
- Pressure adjustment history
- Temperature-corrected pressure calculations

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Pressure change per 10°C	~1 PSI	1-2 PSI	>2 PSI
Winter pressure drop	2-4 PSI below summer	4-6 PSI	>6 PSI
Pressure after temp drop	>90% spec	85-90% spec	<85% spec

Early Warning Signs:
- TPMS warning appearing in cold weather mornings
- Pressure readings 3-5 PSI lower in winter vs. summer
- Pressure fluctuating significantly with weather changes
- Need to adjust pressure monthly during seasonal transitions
- All tires showing similar pressure drop (indicates temperature, not leak)

Preventive Suggestions for Drivers:
- Check and adjust tire pressure monthly, especially during seasonal changes
- Expect ~1 PSI drop per 10°F (5.5°C) temperature decrease
- Inflate tires to proper pressure in cold conditions (morning, before driving)
- Don't reduce pressure in summer if properly inflated in winter
- Use nitrogen inflation for more stable pressure (less affected by temperature)
- Set pressure 2-3 PSI higher in fall to compensate for winter drop
- Monitor TPMS closely during first cold snap of season
- Understand that pressure rises 2-4 PSI during driving (normal)

When Service is Required:
- **Immediate**: Pressure <85% spec after temperature drop
- **Within 1 week**: Pressure 85-90% spec, TPMS warning persistent
- **Routine adjustment**: Seasonal pressure adjustment needed, no damage
- **No service needed**: Pressure fluctuations match temperature changes (normal)

5.14 Combined Predictive Scenario: Multi-Factor Tire Risk Index

Description: AI model combines multiple parameters to calculate comprehensive Tire Health Score or Failure Risk Index.

Data Inputs Required:

- TPMS (pressure + temperature) real-time and historical
- Accelerometer/vibration data
- Vehicle speed, load, alignment data
- Maintenance logs (rotation, balance, replacement history)
- Road surface condition (from suspension/IMU or HD map data)
- Weather and climate data
- Tire age and mileage
- Wear rate calculations

Risk Index Calculation:

Risk Level	Score	Indicators	Action Required
Low Risk	0-30	All parameters normal	Routine monitoring
Moderate Risk	31-60	1-2 warning indicators	Increased monitoring, service within 1 month
High Risk	61-85	3+ warning indicators	Service within 1 week
Critical Risk	86-100	Critical thresholds exceeded	Immediate service/replacement

Composite Risk Factors:

- Pressure deviation from spec (weight: 20%)
- Temperature anomalies (weight: 15%)
- Age and mileage (weight: 15%)
- Wear rate and pattern (weight: 15%)
- Vibration and balance (weight: 10%)
- Load history (weight: 10%)
- Environmental exposure (weight: 10%)
- Maintenance compliance (weight: 5%)

Preventive Suggestions for Drivers:

- Monitor Tire Health Score in vehicle app or telematics dashboard
- Respond to "Tire Health Low" alerts within recommended timeframe
- Maintain consistent inspection and maintenance schedules
- Use OEM mobile apps for real-time tire monitoring
- Set up automatic alerts for pressure, temperature, and risk score changes
- Review detailed tire health reports monthly
- Address multiple warning indicators promptly (compound risk)
- Keep maintenance records updated for accurate risk assessment

When Service is Required:

- **Immediate**: Risk score >85, multiple critical indicators
- **Within 1 week**: Risk score 70-85, high-risk classification
- **Within 2 weeks**: Risk score 60-70, multiple warning indicators
- **Within 1 month**: Risk score 40-60, moderate risk
- **Next scheduled service**: Risk score 30-40, preventive maintenance

SECTION 6: ENERGY MANAGEMENT SYSTEM MONITORING

The Energy Management System (EMS) coordinates power flow throughout the vehicle, optimizing efficiency and performance. Key functions include:

- **Power distribution**: Managing energy flow between battery, motor, and auxiliary systems
- **Regenerative braking control**: Converting kinetic energy back to electrical energy
- **Efficiency optimization**: Balancing performance with energy consumption
- **Range prediction**: Calculating remaining distance based on driving patterns and conditions
- **Thermal management coordination**: Optimizing component temperatures for efficiency
- **Auxiliary power management**: Controlling HVAC, lighting, and accessory loads

Energy Management Failure Prediction Scenarios

6.1 Power Distribution Inefficiency

Description: Energy losses in power distribution due to wiring resistance, contactor degradation, or control algorithm issues.

Data Inputs Required:

- Battery power output (kW)
- Motor power input (kW)
- Auxiliary system power consumption
- High-voltage wiring temperature
- Contactor operation logs
- Power conversion efficiency
- DC-DC converter efficiency

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Distribution efficiency	>97%	94-97%	<94%

Power loss	<3% of total	3-6%	>6%
Wiring temperature rise	<15°C above ambient	15-30°C	>30°C
Contactor resistance	<1mΩ	1-3mΩ	>3mΩ
DC-DC converter efficiency	>90%	85-90%	<85%

Early Warning Signs:
- Energy consumption increasing without driving pattern changes
- High-voltage cables feeling warm to touch (when safe to check)
- Reduced overall vehicle efficiency (kWh/km or kWh/mile increasing)
- Power delivery feeling sluggish despite adequate battery SOC
- Contactor clicking or chattering sounds
- "Power System Fault" warnings
- Auxiliary systems (12V) experiencing voltage drops

Preventive Suggestions for Drivers:
- Monitor energy consumption trends weekly
- Compare kWh/km (or kWh/mile) with historical baseline
- Document efficiency changes with driving conditions
- Report gradual efficiency degradation (>5% over 3 months)
- Ensure all high-voltage connectors are secure during service
- Avoid aftermarket modifications to electrical systems
- Keep vehicle software updated for efficiency optimizations

When Service is Required:
- **Immediate**: Distribution efficiency <92%, burning smell, excessive heat
- **Within 1 week**: Efficiency 92-94%, power loss >5%, persistent warnings
- **Within 1 month**: Efficiency 94-96%, gradual degradation trend
- **Next scheduled service**: Efficiency 96-97%, minor optimization needed

6.2 Regenerative Braking Degradation

Description: Reduced regenerative braking capability due to battery limitations, inverter issues, or control system faults.

Data Inputs Required:
- Regenerative braking power (kW)
- Brake pedal position vs. regen power correlation
- Battery SOC during braking events
- Battery temperature during regen
- Motor/inverter temperature
- Friction brake usage frequency
- Energy recovery efficiency

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Max regen power	90-100% of rated	70-90%	<70%
Regen efficiency	>85%	75-85%	<75%
Energy recovery rate	>20% of total energy	15-20%	<15%
Regen consistency	±5% variation	±5-15%	>±15%
Friction brake usage	<30% of braking	30-50%	>50%

Early Warning Signs:

- Regenerative braking feeling weaker or less responsive
- Friction brakes engaging more frequently during normal braking
- Reduced "one-pedal driving" effectiveness
- Battery not charging as expected during deceleration
- "Regen Limited" or "Brake System" warnings
- Inconsistent braking feel (sometimes strong regen, sometimes weak)
- Energy recovery percentage decreasing over time
- Brake pads wearing faster than expected

Preventive Suggestions for Drivers:

- Monitor regenerative braking power on display during consistent braking events
- Document energy recovery percentage per trip
- Test regen at various SOC levels (limited at high SOC is normal)
- Avoid aggressive braking when battery is very cold (<0°C)
- Allow battery to warm up in cold weather before expecting full regen
- Report sudden changes in regen behavior
- Understand regen limitations: reduced at >85% SOC, in cold weather, at low speeds
- Use friction brakes gently when regen is limited (don't force it)

When Service is Required:

- **Immediate**: Regen power <60% rated, brake system warnings, safety concern
- **Within 1 week**: Regen power 60-75%, efficiency <75%, inconsistent operation
- **Within 1 month**: Regen power 75-85%, energy recovery <18%
- **Next scheduled service**: Regen power 85-90%, minor efficiency reduction

6.3 Range Prediction Accuracy Degradation

Description: Range estimation becoming increasingly inaccurate due to battery capacity loss, efficiency changes, or algorithm errors.

Data Inputs Required:

- Displayed range at various SOC levels
- Actual distance driven vs. range consumed
- Historical driving patterns and efficiency
- Battery capacity (SOH)
- Temperature effects on range
- Auxiliary power consumption

- Terrain and driving style data
- HVAC usage patterns

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Range accuracy	±10% of actual	±10-20%	>±20%
Range prediction stability	±5% variation	±5-15%	>±15%
Displayed range at 100% SOC	95-105% of EPA/WLTP	85-95% or 105-115%	<85% or >115%
Range consumption rate	0.9-1.1 mi/mi displayed	0.8-0.9 or 1.1-1.3	<0.8 or >1.3

Early Warning Signs:

- Range dropping faster than distance traveled (e.g., 10 miles driven, 15 miles range lost)
- Range estimate jumping significantly during driving
- Displayed range at 100% SOC decreasing over time (beyond capacity loss)
- Range prediction not adjusting for driving conditions
- "Guess-O-Meter" becoming unreliable for trip planning
- Range estimate resetting dramatically after charging
- Significant discrepancy between trip computer efficiency and range calculation

Preventive Suggestions for Drivers:

- Track actual range vs. displayed range for consistent trips
- Document range at 100% SOC monthly to identify trends
- Calculate actual efficiency: distance driven ÷ kWh consumed
- Reset trip computer periodically to recalibrate
- Understand factors affecting range: temperature, speed, HVAC, terrain
- Use percentage SOC rather than range for critical planning
- Perform BMS calibration: full charge (100%) followed by deep discharge (10%)
- Report range prediction errors >20% consistently

When Service is Required:

- **Immediate**: Range prediction >30% inaccurate, safety concern for trip planning
- **Within 1 week**: Range prediction 20-30% inaccurate, algorithm suspected fault
- **Within 1 month**: Range prediction 15-20% inaccurate, BMS calibration needed
- **Next scheduled service**: Range prediction 10-15% inaccurate, minor recalibration

6.4 Auxiliary System Power Drain

Description: Excessive power consumption by auxiliary systems (HVAC, lights, infotainment) reducing range and efficiency.

Data Inputs Required:

- 12V battery voltage and current draw
- DC-DC converter load

- HVAC power consumption
- Infotainment system power draw
- Lighting system power
- Parasitic drain when parked
- Auxiliary system efficiency

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
HVAC power consumption	1-4 kW (climate dependent)	4-6 kW	>6 kW
Parasitic drain (parked)	<50W	50-150W	>150W
12V battery drain rate	<5% per week parked	5-15% per week	>15% per week
DC-DC converter efficiency	>85%	75-85%	<75%
Total aux power	<10% of drive power	10-20%	>20%

Early Warning Signs:

- Range significantly reduced when using HVAC (>30% impact)
- 12V battery dying when parked for 1-2 weeks
- Infotainment or lights draining battery noticeably
- "12V Battery Low" warnings
- Auxiliary systems not shutting off when vehicle locked
- Excessive power draw visible on energy flow display
- Range loss while parked (phantom drain)

Preventive Suggestions for Drivers:

- Pre-condition cabin while plugged in (uses grid power, not battery)
- Use seat heaters instead of cabin heat when possible (more efficient)
- Minimize HVAC use in extreme temperatures
- Turn off unnecessary accessories (heated seats, defrosters when not needed)
- Use "Camp Mode" or "Dog Mode" efficiently when parked with systems on
- Ensure all lights and accessories off when exiting vehicle
- Monitor 12V battery health (replace every 3-5 years)
- Disconnect 12V battery if storing vehicle >1 month
- Check for aftermarket accessories causing parasitic drain

When Service is Required:

- **Immediate**: 12V battery repeatedly dying, parasitic drain >200W, safety concern
- **Within 1 week**: Parasitic drain 150-200W, 12V battery draining >20%/week
- **Within 1 month**: HVAC power >5 kW consistently, aux systems >15% of drive power
- **Next scheduled service**: 12V battery aging (>4 years), minor parasitic drain

SECTION 7: THERMAL MANAGEMENT SYSTEM MONITORING

The Thermal Management System maintains optimal operating temperatures for all critical components:

- **Battery cooling/heating**: Liquid or air cooling, resistive or heat pump heating
- **Motor and inverter cooling**: Liquid cooling circuits with radiator and pump
- **Cabin HVAC**: Heat pump or resistive heating, air conditioning
- **Coolant pumps**: Circulating coolant through components
- **Radiators and heat exchangers**: Dissipating heat to ambient air
- **Coolant reservoir and sensors**: Monitoring fluid level and temperature
- **Thermal control valves**: Directing coolant flow to components as needed

Thermal Management Failure Prediction Scenarios

7.1 Cooling System Pump Failure

Description: Coolant pump degradation or failure reducing flow rate and cooling capacity.

Data Inputs Required:

- Coolant pump speed (RPM)
- Coolant flow rate
- Pump power consumption
- Coolant inlet/outlet temperature delta
- Component temperatures (battery, motor, inverter)
- Pump vibration/noise (if sensors available)
- Pump run time and cycles

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Pump flow rate	90-100% of rated	70-90%	<70%
Pump speed	90-100% commanded	70-90%	<70%
Power consumption	50-150W (typical)	150-200W or <30W	>200W or <30W
Coolant ΔT	5-15°C	15-25°C	>25°C
Component temp rise rate	<2°C/min	2-4°C/min	>4°C/min

Early Warning Signs:

- Coolant temperature delta increasing (reduced flow)
- Component temperatures rising faster under load
- Pump noise (grinding, squealing, rattling)

- Pump running at higher speeds to maintain cooling
- "Cooling System Fault" warnings
- Coolant temperature rising during normal driving
- Reduced performance due to thermal limiting
- Pump power consumption abnormal (too high or too low)

Preventive Suggestions for Drivers:

- Listen for pump noise during vehicle operation (quiet hum is normal)
- Monitor component temperatures during consistent driving
- Check coolant level monthly (low level can damage pump)
- Avoid running vehicle with low coolant
- Service cooling system per manufacturer schedule
- Report cooling system warnings immediately
- Reduce power demands if thermal limiting occurs
- Allow vehicle to cool if temperatures elevated

When Service is Required:

- **STOP IMMEDIATELY**: Pump failure, coolant leak, component temp >critical threshold
- **Immediate**: Pump flow <75%, persistent cooling warnings, component temps elevated
- **Within 1 week**: Pump noise, flow 75-85%, coolant ΔT >20°C
- **Next scheduled service**: Minor pump noise, flow 85-90%, gradual degradation

7.2 Radiator or Heat Exchanger Blockage

Description: Reduced heat dissipation due to debris, corrosion, or internal blockage in radiator or heat exchanger.

Data Inputs Required:

- Coolant inlet/outlet temperature
- Radiator fan speed
- Ambient temperature
- Vehicle speed (ram air effect)
- Heat rejection capacity
- Coolant flow rate
- Component temperatures

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Heat rejection capacity	90-100% of design	70-90%	<70%
Coolant temp at given load	Design temp ±5°C	Design +5-15°C	>Design +15°C
Fan duty cycle	<50% at moderate ambient	50-80%	>80%
Radiator effectiveness	>85%	70-85%	<70%

Early Warning Signs:

- Coolant temperature higher than normal for given conditions

- Radiator fan running continuously or at high speed
- Component temperatures elevated despite adequate coolant flow
- Reduced cooling capacity in hot weather
- Thermal limiting occurring more frequently
- Visible debris or blockage in radiator fins
- Coolant temperature not dropping during highway driving (ram air should help)

Preventive Suggestions for Drivers:
- Inspect radiator fins for debris, leaves, plastic bags
- Clean radiator exterior with low-pressure water (not high-pressure)
- Avoid driving through tall grass or debris fields
- Monitor coolant temperature in hot weather
- Ensure adequate airflow to radiator (no aftermarket modifications blocking)
- Service cooling system flush per schedule (removes internal deposits)
- Report gradual temperature increases over weeks/months

When Service is Required:
- **Immediate**: Heat rejection <70%, component temps critical, severe blockage visible
- **Within 1 week**: Heat rejection 70-80%, coolant temps elevated 10-15°C above normal
- **Within 1 month**: Heat rejection 80-85%, fan running excessively
- **Next scheduled service**: Heat rejection 85-90%, minor blockage, preventive cleaning

7.3 Coolant Leak or Low Fluid Level

Description: Coolant loss due to leaks, evaporation, or system failure, reducing cooling capacity and risking component damage.

Data Inputs Required:
- Coolant level sensor
- Coolant consumption rate
- Component temperatures
- Coolant pressure
- Refill frequency logs
- Leak detection sensors (if available)
- Coolant temperature trends

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Coolant level	90-100% full	70-90%	<70%
Coolant loss rate	<50ml per 10,000 km	50-200ml per 10,000 km	>200ml per 10,000 km
Refill frequency	<1x per year	1-3x per year	>3x per year
System pressure	1.0-1.5 bar	0.7-1.0 bar	<0.7 bar

Component temp increase	Baseline	+5-10°C	>+10°C

Early Warning Signs:
- "Coolant Level Low" warning on dashboard
- Visible coolant puddles under vehicle after parking
- Sweet smell (ethylene glycol) near vehicle
- Coolant level requiring frequent top-ups
- Component temperatures gradually increasing
- White steam from engine compartment
- Coolant stains on components or ground
- Pressure loss in cooling system

Preventive Suggestions for Drivers:
- Check coolant level monthly when vehicle cold
- Inspect under vehicle for leaks after parking overnight
- Look for coolant stains on driveway/parking spot
- Document coolant additions (date, amount, mileage)
- Use only manufacturer-specified coolant type
- Never open cooling system when hot (severe burn risk)
- Address small leaks immediately—they worsen rapidly
- Inspect hoses and connections during service for wear/cracks

When Service is Required:
- **STOP IMMEDIATELY**: Rapid coolant loss, steam, component temps critical
- **Do not drive**: Coolant level <50%, active leak visible, overheating
- **Immediate**: Coolant level 50-70%, persistent leak, frequent refills needed
- **Within 1 week**: Coolant level 70-80%, slow leak detected, refills monthly
- **Next scheduled service**: Coolant level 80-90%, minor seepage, annual refill

7.4 Coolant Quality Degradation

Description: Coolant loses protective properties due to age, contamination, or chemical breakdown, reducing heat transfer and causing corrosion.

Data Inputs Required:
- Coolant age (time since last change)
- Coolant pH level (if sensors available)
- System corrosion indicators
- Heat transfer efficiency
- Coolant temperature stability
- Pump and component condition
- Mileage since coolant service

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Coolant age	<5 years or 150,000 km	5-8 years or 150-200k km	>8 years or >200k km
pH level	7.5-11.0	7.0-7.5 or 11.0-11.5	<7.0 or >11.5
Heat transfer efficiency	>95% of new	85-95%	<85%
Corrosion indicators	None	Minor deposits	Significant corrosion
Coolant appearance	Clear, proper color	Slightly discolored	Dark, contaminated

Early Warning Signs:

- Coolant appearing rusty, brown, or contaminated
- Coolant smelling acidic or unusual
- Reduced cooling efficiency over time
- Corrosion visible on metal components
- Coolant system deposits or sludge
- pH outside acceptable range (test strips available)
- Coolant hasn't been changed per schedule

Preventive Suggestions for Drivers:

- Follow manufacturer coolant change intervals (typically 5-8 years or 150,000 km)
- Use only specified coolant type (don't mix different types)
- Inspect coolant color and clarity annually
- Have coolant tested at service appointments
- Flush system completely when changing coolant (not just drain and fill)
- Keep records of coolant service dates
- Consider earlier replacement in severe climates or heavy use

When Service is Required:

- **Immediate**: Coolant dark/contaminated, pH <7.0 or >11.5, visible corrosion
- **Within 1 month**: Coolant >8 years old, heat transfer efficiency <85%
- **Within 3 months**: Coolant 5-8 years old, pH 7.0-7.5, minor discoloration
- **Next scheduled service**: Approaching service interval, preventive replacement

7.5 Thermal Control Valve Malfunction

Description: Valves controlling coolant flow to different components fail to operate properly, causing uneven cooling or overheating.

Data Inputs Required:

- Valve position sensors
- Coolant flow distribution
- Component temperatures (battery, motor, inverter individually)
- Valve command vs. actual position
- Temperature response to valve commands
- Valve cycling frequency

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Valve position accuracy	±5% of commanded	±5-15%	>±15%
Valve response time	<5 seconds	5-15 seconds	>15 seconds
Component temp deviation	<10°C between components	10-20°C	>20°C
Valve cycling	Smooth operation	Occasional sticking	Frequent sticking/failure

Early Warning Signs:

- One component overheating while others normal temperature
- Uneven temperature distribution (e.g., battery hot, motor cool)
- Slow temperature response to driving conditions
- "Thermal System Fault" warnings
- Component thermal limiting despite adequate cooling capacity
- Valve clicking or unusual sounds
- Temperature swings (cycling hot and cold)

Preventive Suggestions for Drivers:

- Monitor individual component temperatures if available
- Note temperature response patterns during driving
- Report uneven cooling (one component hot, others cool)
- Listen for valve operation sounds (clicking during startup normal)
- Avoid ignoring thermal system warnings
- Document temperature patterns for service diagnosis

When Service is Required:

- **Immediate**: Component overheating due to valve failure, safety concern
- **Within 1 week**: Valve position error >10%, component temp deviation >15°C
- **Within 1 month**: Valve response slow (>10 sec), occasional sticking
- **Next scheduled service**: Minor valve position errors, preventive inspection

7.6 Battery Thermal Management Failure

Description: Battery heating or cooling system fails to maintain optimal temperature range, affecting performance, charging, and lifespan.

Data Inputs Required:

- Battery pack temperature (multiple sensors)
- Battery heater power consumption
- Battery cooling system status
- Ambient temperature
- Charging rate limitations due to temperature
- Power limitations due to temperature

- Pre-conditioning effectiveness

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Battery temp (optimal)	15-35°C (59-95°F)	5-15°C or 35-45°C	<5°C or >45°C
Heating/cooling effectiveness	Target temp in <30 min	30-60 min	>60 min
Temp uniformity	<5°C across pack	5-10°C	>10°C
Pre-conditioning power	2-6 kW	1-2 kW or 6-8 kW	<1 kW or >8 kW

Early Warning Signs:
- Battery remaining cold in winter despite pre-conditioning
- Battery overheating in summer despite cooling
- Reduced charging rate due to temperature ("Charging Slow - Battery Temp")
- Reduced power delivery in cold weather (more than expected)
- Pre-conditioning taking longer than normal
- Uneven battery temperatures across modules
- Battery heater or cooler not activating

Preventive Suggestions for Drivers:
- Pre-condition battery before driving in extreme temperatures
- Pre-condition while plugged in (uses grid power, not battery)
- Allow adequate pre-conditioning time (15-30 minutes)
- Park in garage during extreme weather when possible
- Schedule charging to complete before departure (battery warm)
- Monitor battery temperature before fast charging
- Avoid fast charging with very cold battery (<5°C)
- Report pre-conditioning failures immediately

When Service is Required:
- **Immediate**: Battery temp >45°C or <0°C with system failure, safety concern
- **Within 1 week**: Pre-conditioning ineffective, temp uniformity >8°C
- **Within 1 month**: Heating/cooling taking >45 min, power consumption abnormal
- **Next scheduled service**: Minor pre-conditioning delays, temp uniformity 5-8°C

SECTION 8: HIGH VOLTAGE SAFETY SYSTEM MONITORING

The High Voltage Safety System protects occupants and technicians from electrical hazards:
- **Insulation monitoring**: Continuously measures resistance between HV system and chassis
- **Ground fault detection**: Identifies current leakage to ground
- **Interlock system**: Safety switches that disconnect HV when opened
- **Manual service disconnect**: Allows technicians to safely de-energize HV system

- **Crash sensors**: Automatically disconnect HV in collision
- **Arc fault detection**: Identifies dangerous electrical arcing
- **HV contactors**: High-current relays that connect/disconnect battery

High Voltage Safety Failure Prediction Scenarios

8.1 Insulation Resistance Degradation

Description: Electrical insulation between high-voltage system and vehicle chassis degrades, creating shock hazard and reducing safety margins.

Data Inputs Required:
- Insulation resistance measurements (Ω)
- Leakage current to chassis
- Environmental humidity
- Component temperatures
- HV system voltage
- Insulation monitoring system status
- Historical resistance trends

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Insulation resistance	>500 Ω/V (>200 MΩ @ 400V)	100-500 Ω/V (40-200 MΩ)	<100 Ω/V (<40 MΩ)
Leakage current	<0.5 mA	0.5-2 mA	>2 mA
Resistance trend	Stable	Decreasing 10-30%/year	Decreasing >30%/year
Humidity correlation	Minimal effect	Moderate effect	Strong effect

Early Warning Signs:
- "Insulation Fault" or "High Voltage System Fault" warnings
- Insulation resistance decreasing over weeks/months
- Leakage current increasing
- Resistance drops significantly in wet/humid conditions
- Tingling sensation when touching vehicle (DANGER - do not ignore)
- Electrical smell from HV components
- Insulation monitoring system warnings

Preventive Suggestions for Drivers:
- **DO NOT DRIVE** if "Insulation Fault" warning appears
- Avoid driving through deep water or flooded roads
- Keep HV components dry (no high-pressure washing of underbody)

- Park in covered areas during extended wet weather
- Report any tingling or shock sensations immediately
- Have insulation tested during annual service
- Address coolant leaks near HV components immediately
- Avoid aftermarket modifications to HV system

When Service is Required:
- **DO NOT DRIVE**: Insulation resistance <100 Ω/V, leakage current >2 mA, shock hazard
- **Immediate**: Insulation resistance 100-200 Ω/V, persistent warnings
- **Within 1 week**: Insulation resistance 200-300 Ω/V, leakage 1-2 mA
- **Within 1 month**: Insulation resistance 300-400 Ω/V, decreasing trend
- **Next scheduled service**: Insulation resistance 400-500 Ω/V, preventive inspection

8.2 Ground Fault Detection

Description: Current leaking from HV system to chassis ground, indicating insulation failure or component damage.

Data Inputs Required:
- Ground fault current magnitude
- Fault location (if system can isolate)
- Fault persistence (intermittent vs. continuous)
- Environmental conditions during fault
- Component temperatures
- Recent service or damage history

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Ground fault current	0 mA	0.5-2 mA	>2 mA
Fault frequency	Never	1-5 per month	>5 per month
Fault persistence	N/A	Clears within minutes	Persistent

Early Warning Signs:
- "Ground Fault Detected" warning
- Vehicle refusing to enter "Ready" mode
- HV system automatically disconnecting
- Intermittent power loss
- Fault occurring in wet conditions
- Recent water exposure or damage
- Coolant leak near HV components

Preventive Suggestions for Drivers:
- **DO NOT DRIVE** with active ground fault warning
- **DO NOT TOUCH** vehicle if ground fault suspected
- Avoid water exposure to HV components

- Report ground fault warnings immediately even if intermittent
- Document conditions when fault occurs (weather, location, recent events)
- Have vehicle towed to service facility (do not drive)
- Ensure only certified HV technicians work on vehicle

When Service is Required:
- **DO NOT DRIVE**: Any ground fault detected, immediate towing required
- **Emergency**: Persistent ground fault, shock hazard, safety critical
- **All ground faults require immediate professional service**

8.3 Interlock System Failure

Description: Safety interlock switches fail to properly disconnect HV system when service panels opened or in crash.

Data Inputs Required:
- Interlock switch status (all switches)
- Interlock circuit continuity
- Service panel position sensors
- Crash sensor status
- HV contactor status
- Interlock system self-test results

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Interlock continuity	Complete circuit	Intermittent	Open circuit
Switch operation	100% reliable	95-100%	<95%
Response time	<100 ms	100-500 ms	>500 ms
Self-test pass rate	100%	95-100%	<95%

Early Warning Signs:
- "Interlock System Fault" warning
- HV system not disconnecting when service panel opened
- Intermittent interlock warnings
- Vehicle entering "Ready" mode with panel open
- Interlock self-test failures
- Corrosion or damage to interlock connectors

Preventive Suggestions for Drivers:
- **NEVER** open HV service panels (only certified technicians)
- Report interlock warnings immediately
- Ensure service panels fully closed and latched
- Inspect interlock connectors during service for corrosion
- Do not bypass or defeat interlock systems (illegal and dangerous)

- Have interlock system tested during annual service

When Service is Required:
- **DO NOT DRIVE**: Interlock system failure, HV not disconnecting when required
- **Immediate**: Interlock warnings, intermittent operation, self-test failures
- **Within 1 week**: Interlock connector corrosion, minor issues
- **Next scheduled service**: Preventive interlock system inspection

8.4 HV Contactor Degradation

Description: Main HV contactors (relays) degrade due to arcing, contact wear, or coil failure, affecting HV connection reliability.

Data Inputs Required:
- Contactor operation cycles
- Contactor resistance when closed
- Contactor opening/closing time
- Voltage drop across contacts
- Contactor temperature
- Arc detection events
- Welding detection (contacts stuck closed)

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Contact resistance	<1 mΩ	1-3 mΩ	>3 mΩ
Operation cycles	<100,000	100,000-150,000	>150,000
Voltage drop	<0.5V at rated current	0.5-1.5V	>1.5V
Opening/closing time	<50 ms	50-100 ms	>100 ms
Welding events	0	1-2 lifetime	>2

Early Warning Signs:
- Clicking or chattering from HV contactors
- Delayed vehicle "Ready" mode activation
- "HV System Fault" warnings
- Contactors not opening (vehicle won't shut down)
- Contactors not closing (vehicle won't start)
- Burning smell from contactor area
- Voltage drop increasing over time
- Arc flash visible or audible during operation
- Vehicle refusing to charge (contactors not closing)

Preventive Suggestions for Drivers:
- Allow vehicle to fully shut down before exiting (don't force power off)

- Avoid repeatedly cycling vehicle on/off rapidly
- Report clicking or unusual sounds from HV area
- Document delayed startup or shutdown issues
- Monitor contactor operation cycles if displayed
- Have contactors inspected during major service intervals
- Report any "HV System" warnings immediately

When Service is Required:
- **DO NOT DRIVE**: Contactors stuck closed (won't shut down), welding detected
- **Immediate**: Contactors stuck open (won't start), persistent clicking, burning smell
- **Within 1 week**: Contact resistance >2 mΩ, voltage drop >1V, arc events
- **Within 1 month**: Contact resistance 1-2 mΩ, >100,000 cycles, slow operation
- **Next scheduled service**: Approaching cycle limit, preventive inspection

8.5 Arc Fault Detection Events

Description: Dangerous electrical arcing detected in HV system, indicating damaged insulation, loose connections, or component failure.

Data Inputs Required:
- Arc fault detection events (frequency, location)
- Current waveform anomalies
- Voltage transients
- Component temperatures
- Recent service or damage history
- Environmental conditions during events

Threshold Ranges:

Parameter	Normal Range	Warning Range	Critical Range
Arc events	0 per month	1-3 per month	>3 per month
Arc duration	N/A	<100 ms	>100 ms
Arc current	N/A	<50A	>50A
Event persistence	N/A	Intermittent	Repeated/continuous

Early Warning Signs:
- "Arc Fault Detected" warning
- Popping or crackling sounds from HV components
- Ozone smell (sharp, electrical smell)
- Visible sparking or flashing
- Intermittent power loss
- HV system automatically disconnecting
- Burn marks or discoloration on components

Preventive Suggestions for Drivers:

- **STOP IMMEDIATELY** if arc fault detected
- **DO NOT DRIVE** with arc fault warnings
- Report any electrical smells or sounds immediately
- Avoid water exposure to HV components
- Ensure all HV connectors secure during service
- Document conditions when arc events occur
- Have vehicle towed to service facility

When Service is Required:

- **DO NOT DRIVE**: Any arc fault detected, immediate towing required
- **Emergency**: Persistent arcing, visible sparks, fire hazard
- **All arc faults require immediate professional service**

SECTION 9: COMPREHENSIVE MONITORING SCHEDULE

Daily Checks (Before Each Drive)

Visual Inspection (2 minutes)

- ✓ Tire pressure warning lights (TPMS)
- ✓ Dashboard warning lights or messages
- ✓ Visible tire damage, low pressure, or unusual wear
- ✓ Fluid puddles under vehicle (coolant, brake fluid)
- ✓ Charging port condition (if charging overnight)
- ✓ 12V battery status (if displayed)

Functional Check (during first 5 minutes of driving)

- ✓ Regenerative braking response
- ✓ Acceleration performance normal
- ✓ No unusual sounds (grinding, clicking, whining)
- ✓ No unusual vibrations
- ✓ HVAC and auxiliary systems functioning
- ✓ Range estimate reasonable for SOC level

Action Items:

- Address any warning lights before driving
- Document unusual symptoms for trend analysis
- Check tire pressure if TPMS warning appears

Weekly Checks (5-10 minutes)

Tire System

- ✓ Manual tire pressure check (all 4 tires + spare)
- ✓ Visual inspection for embedded objects, cuts, bulges
- ✓ Tread depth spot check (penny test: 2/32" minimum)
- ✓ Uneven wear patterns

Charging System

- ✓ Charging port cleanliness (wipe with dry cloth)
- ✓ Charging cable condition (no damage, corrosion)
- ✓ Consistent charging times for similar sessions
- ✓ Charging power delivery as expected

Energy Management

- ✓ Energy consumption trends (kWh/km or kWh/mile)
- ✓ Regenerative braking consistency
- ✓ Range estimate accuracy
- ✓ Auxiliary power consumption patterns

Thermal Management

- ✓ Component temperatures during normal driving
- ✓ Cooling fan operation (should be quiet, not constant)
- ✓ No overheating warnings

Action Items:

- Inflate tires to specification if low
- Clean charging port contacts if dirty
- Document consumption trends for baseline comparison
- Report any gradual changes in performance

Monthly Checks (15-20 minutes)

Battery System

- ✓ Perform full charge cycle (20% → 100%) for BMS calibration
- ✓ Document displayed range at 100% SOC
- ✓ Check for cell voltage imbalance warnings
- ✓ Review charging logs for anomalies
- ✓ Battery temperature behavior after driving

Motor and Drivetrain

- ✓ Listen for new or changing motor sounds
- ✓ Consistent acceleration test (0-60 km/h or 0-40 mph)

- ✓ Motor temperature after spirited driving
- ✓ No vibration or unusual sensations

Thermal Management
- ✓ Coolant level check (when cold)
- ✓ Visual inspection for leaks under vehicle
- ✓ Coolant appearance (should be clear, proper color)
- ✓ No coolant stains on components

High Voltage Safety
- ✓ No HV system warnings or faults
- ✓ No tingling or shock sensations
- ✓ Service panel latches secure
- ✓ No electrical smells

Efficiency Baseline
- ✓ Calculate actual efficiency: distance ÷ kWh consumed
- ✓ Compare to previous month and historical baseline
- ✓ Document any efficiency changes >5%

Action Items:
- Top up coolant if low (use specified type only)
- Address any new sounds or vibrations
- Report efficiency degradation >5%
- Schedule service if warnings appear

Quarterly Checks (30-45 minutes)

Comprehensive Battery Assessment
- ✓ Capacity test: 100% → 10%, record kWh consumed
- ✓ Compare capacity to manufacturer spec and previous tests
- ✓ Cell voltage deviation check (if accessible)
- ✓ Battery temperature uniformity
- ✓ Charging efficiency calculation

Motor System Detailed Check
- ✓ Motor temperature during 30-minute highway drive
- ✓ Compare performance metrics with baseline
- ✓ Vibration assessment at various speeds
- ✓ Bearing noise check (listen during coasting)
- ✓ Motor compartment visual inspection

Inverter and Power Electronics
- ✓ Inverter temperature during consistent driving

- ✓ Regenerative braking power and consistency
- ✓ Phase balance (if data available)
- ✓ Efficiency metrics vs. baseline

Tire Comprehensive Inspection
- ✓ Tread depth measurement (all positions)
- ✓ Wear pattern analysis (alignment indicators)
- ✓ Tire age check (DOT date code)
- ✓ Sidewall condition (cracks, bulges, damage)
- ✓ Tire rotation (if due per schedule)

Thermal System Assessment
- ✓ Coolant condition (color, clarity, level)
- ✓ Cooling system pressure test (if equipment available)
- ✓ Heat rejection capacity during load
- ✓ Pre-conditioning effectiveness test
- ✓ Component temperature uniformity

Energy Management Review
- ✓ 3-month energy consumption trend analysis
- ✓ Range prediction accuracy assessment
- ✓ Regenerative braking efficiency
- ✓ Auxiliary power consumption patterns
- ✓ HVAC efficiency in various conditions

Documentation and Trending
- ✓ Update maintenance log with all measurements
- ✓ Create trend charts for key parameters
- ✓ Compare current data to 3, 6, 12 months prior
- ✓ Identify any degradation patterns
- ✓ Plan service appointments based on trends

Action Items:
- Schedule service for any parameters in warning range
- Replace tires if tread <4/32" or age >6 years
- Flush coolant if due per schedule
- Update baseline measurements for future comparison

Annual Checks (Professional Service Recommended)

Comprehensive Professional Inspection
- ✓ High-voltage system insulation resistance test
- ✓ Battery pack detailed diagnostics
- ✓ Motor bearing inspection and lubrication check

- ✓ Inverter thermal imaging and electrical testing
- ✓ Cooling system pressure test and flush (if due)
- ✓ Brake system inspection (pads, fluid, calipers)
- ✓ Suspension and alignment check
- ✓ Tire rotation and balance
- ✓ 12V battery load test and replacement (if >4 years)
- ✓ Software and firmware updates
- ✓ Comprehensive diagnostic scan
- ✓ Safety system verification (interlocks, ground fault)

Preventive Replacements (Time/Mileage Based)

- ✓ Cabin air filter (annually or per schedule)
- ✓ Brake fluid (every 2-3 years)
- ✓ Coolant (every 5-8 years or per manufacturer)
- ✓ 12V battery (every 3-5 years)
- ✓ Wiper blades (as needed)
- ✓ Tires (when tread <4/32" or age >6 years)

SECTION 10: EMERGENCY PROCEDURES

10.1 Battery Thermal Event Response

Recognition Signs:

- Battery temperature >60°C (140°F) with rapid rise
- Smoke or flames from battery area
- Hissing, popping, or venting sounds
- Strong chemical smell
- "Battery Thermal Event" warning
- Visible swelling or deformation of battery pack

IMMEDIATE ACTIONS (Do not delay):

Step 1: Stop Vehicle Safely (10-15 seconds)

1. Pull over to safe location away from buildings, vehicles, and people
2. Turn off vehicle completely
3. Put in Park and engage parking brake
4. Open all windows and doors

Step 2: Evacuate (15-20 seconds)

1. All occupants exit vehicle immediately
2. Move at least 30 meters (100 feet) away

3. Move upwind if smoke present
4. Do not attempt to retrieve belongings

Step 3: Alert and Isolate (30 seconds)

1. Call emergency services immediately: "Electric vehicle battery fire/thermal event"
2. Provide exact location and vehicle description
3. Warn nearby people to evacuate area
4. Establish 30-meter (100-foot) perimeter
5. Do not allow anyone to approach vehicle

Step 4: Monitor and Report (ongoing)

1. Watch for fire, smoke, or venting from safe distance
2. Note wind direction and move further if smoke approaches
3. Inform emergency responders:
    - Vehicle make, model, and approximate battery size
    - Location of battery pack (usually under floor)
    - Any visible fire or smoke
    - Whether vehicle was charging
4. Do not approach vehicle until fire department declares safe

⚠ **CRITICAL SAFETY WARNINGS**:

- **DO NOT** attempt to fight battery fire with standard extinguisher
- **DO NOT** open hood or approach battery area
- **DO NOT** use water unless trained firefighter (special protocols required)
- **DO NOT** re-enter vehicle for any reason
- **Battery fires can reignite hours or days later** - vehicle must be monitored

Post-Event Actions:

1. Vehicle must be towed to specialized facility (not standard tow yard)
2. Vehicle should be placed in quarantine area away from structures
3. Battery must be monitored for 24-48 hours for re-ignition
4. Do not attempt to charge or operate vehicle
5. Contact manufacturer and insurance immediately

10.2 High Voltage Exposure Protocol

Recognition Signs:

- Tingling or shock sensation when touching vehicle
- "Insulation Fault" or "Ground Fault" warning
- Visible damage to HV components (orange cables/connectors)
- Electrical burning smell
- Person in contact with vehicle unable to let go (muscle contractions)
- Exposed orange HV cables after collision

IF SOMEONE IS IN CONTACT WITH ENERGIZED VEHICLE:

Step 1: Do Not Touch Victim or Vehicle (immediate)

1. **DO NOT TOUCH** the person or vehicle - you will be shocked too
2. Yell for help and call emergency services immediately
3. State: "Person being electrocuted by electric vehicle"

Step 2: Break the Circuit (if safe to do so)

1. Look for non-conductive object (dry wood, rubber, plastic)
2. Use object to push victim away from vehicle
3. **Only attempt if you can do so without touching victim or vehicle**
4. If unsure, wait for emergency responders

Step 3: Provide First Aid (after circuit broken)

1. Check for breathing and pulse
2. Begin CPR if needed and trained
3. Treat for electrical burns (do not apply water)
4. Keep victim warm and still
5. Monitor for shock symptoms

IF YOU RECEIVE SHOCK FROM VEHICLE:

Step 1: Break Contact Immediately

1. Pull hand away forcefully if able
2. Jump away from vehicle with both feet together
3. Do not touch vehicle again

Step 2: Assess Injury

1. Check for burns (entry and exit wounds)
2. Monitor for irregular heartbeat, chest pain
3. Seek medical attention immediately even if feeling okay
4. Electrical injuries can have delayed effects

Step 3: Secure Area

1. Mark vehicle as hazardous
2. Prevent others from touching vehicle
3. Call emergency services
4. Report to manufacturer

COLLISION WITH HV SYSTEM DAMAGE:

Step 1: Assess HV System Integrity

- Look for damaged orange cables or connectors
- Check for "HV System Fault" warnings
- Look for sparking, smoke, or electrical sounds
- Smell for electrical burning

Step 2: If HV Damage Suspected

1. Do not touch vehicle exterior
2. Evacuate all occupants without touching metal
3. Establish 10-meter (30-foot) perimeter
4. Call emergency services: "EV collision with possible HV damage"
5. Wait for trained first responders

Step 3: First Responder Information

1. Inform responders: "Electric vehicle with high-voltage system"
2. Point out orange HV cables if visible
3. Indicate battery location (under floor)
4. Mention any warnings or sounds observed
5. Provide emergency response guide if available (usually in glove box)

⚠ **CRITICAL SAFETY WARNINGS**:

- **High voltage can be LETHAL** (300-800V DC)
- **Orange cables/connectors indicate HV** - never touch
- **Vehicle body can become energized** if insulation fails
- **Wet conditions increase shock risk** dramatically
- **Only certified HV technicians** should work on HV system
- **12V disconnect does NOT de-energize HV system**

10.3 Vehicle Immobilization and Securing

Description: Proper procedures to safely immobilize and secure an EV in emergency situations, preventing movement and further hazards.

When Vehicle Immobilization is Required

- Vehicle involved in collision
- Fire or thermal event
- High voltage system fault with safety concern
- Vehicle unresponsive or systems malfunctioning
- Preparing vehicle for towing or recovery
- Suspected electrical hazard

10.3.1 Standard Immobilization Procedure

Step 1: Bring Vehicle to Complete Stop

1. Apply service brake firmly and bring vehicle to controlled stop
2. Shift transmission to PARK (P) position
3. Engage electronic parking brake
4. Turn off vehicle (press power button or remove key)
5. Verify "Ready" indicator light is OFF

Step 2: Disconnect 12V Battery (If Safe)

1. ⚠ **ONLY if no fire, smoke, or electrical hazards present**
2. Open hood/frunk using emergency release if needed
3. Locate 12V battery (usually in frunk or under rear seat)
4. Disconnect NEGATIVE (-) terminal first
5. Disconnect POSITIVE (+) terminal second
6. Secure terminals away from battery posts
7. **Purpose**: Prevents parasitic drain and disables most vehicle systems

Step 3: Activate Hazard Warning Lights

1. Press hazard light button before disconnecting 12V battery
2. Place warning triangles or flares 50-150 meters behind vehicle
3. Use emergency flashlight or road flares if available
4. Ensure vehicle visible to approaching traffic

Step 4: Prevent Vehicle Movement

1. Place wheel chocks on both sides of at least two wheels
2. If on incline, chock downhill wheels
3. If available, use parking blocks or heavy objects
4. Verify parking brake engaged and holding
5. Ensure transmission in PARK position

Step 5: Secure Vehicle Perimeter

1. Establish safety perimeter of at least 5 meters (15 feet)
2. Keep bystanders away from vehicle
3. Post warning signs if available
4. Do not allow smoking or open flames within 15 meters
5. Prevent unauthorized access to vehicle

10.3.2 Emergency Immobilization (Fire or Electrical Hazard)

⚠ **CRITICAL: If fire, smoke, or electrical hazard present:**

Immediate Actions:

1. **EVACUATE ALL OCCUPANTS IMMEDIATELY**
2. Move at least 15 meters (50 feet) away from vehicle
3. Call emergency services (fire department and police)
4. **DO NOT attempt to disconnect 12V battery**
5. **DO NOT open hood or access panels**
6. Keep all persons away from vehicle

Establish Safety Zone:

1. Create 15-meter (50-foot) minimum safety perimeter
2. Prevent vehicle traffic from approaching
3. Warn approaching vehicles using hazard lights, flares, or flags
4. Position emergency responders upwind of vehicle

5. Identify HV battery location to emergency responders

Monitor Vehicle:

1. Watch for signs of thermal runaway:
    - Smoke or steam from battery area
    - Popping or hissing sounds
    - Flames or sparks
    - Strong chemical odor
    - Battery swelling or deformation
2. Be prepared to expand safety perimeter if situation worsens
3. Do not re-approach vehicle until emergency services clear it

Information for Emergency Responders:

- Vehicle make, model, and year
- Battery capacity (kWh) if known
- Location of HV battery (usually underfloor)
- Location of manual service disconnect (if known)
- Any visible damage to battery or HV components
- Duration and nature of thermal event

10.3.3 Collision Immobilization Procedure

Immediate Post-Collision Actions:

Step 1: Assess Safety

1. Check for injuries to all occupants
2. Check for fire, smoke, or fuel leaks
3. Assess vehicle stability (upright, on side, overturned)
4. Identify immediate hazards (traffic, fire, electrical)
5. Decide: stay in vehicle or evacuate

Step 2: Secure Vehicle (If Safe)

1. Turn off vehicle if still powered
2. Engage parking brake
3. Shift to PARK position
4. **DO NOT disconnect 12V battery** (airbag data may be lost)
5. Activate hazard lights
6. Deploy warning triangles

Step 3: High Voltage Safety Assessment

1. Look for HV warning indicators on dashboard
2. Check for visible damage to battery area (underfloor)
3. Look for coolant leaks (may be pink/orange liquid)
4. Listen for hissing, popping, or electrical sounds
5. Smell for chemical odors or burning

6. **If any HV hazards suspected: EVACUATE and establish 15m perimeter**

Step 4: Document and Report

1. Take photos of vehicle position and damage
2. Note any warning lights or messages
3. Document HV system status if visible
4. Report collision to emergency services
5. Inform responders that vehicle is electric

Step 5: Await Professional Assessment

1. Do not attempt to move vehicle without professional guidance
2. Allow certified technicians to assess HV system safety
3. Follow towing procedures for EVs (see Section 10.3.4)
4. Obtain post-collision HV system inspection before driving

10.3.4 Towing and Recovery Procedures

⚠️ **IMPORTANT: Improper towing can damage EV drivetrain and create hazards**

Flatbed Towing (PREFERRED METHOD):

1. Use flatbed or wheel-lift tow truck only
2. Load vehicle onto flatbed with all four wheels off ground
3. Secure vehicle with wheel straps (not frame or suspension)
4. Ensure parking brake released before loading
5. Disconnect 12V battery if tow will exceed 1 hour
6. Transport in PARK or NEUTRAL (consult owner's manual)

Dolly Towing (IF FLATBED UNAVAILABLE):

1. **Front-Wheel Drive EVs**: Tow with front wheels on dolly
2. **Rear-Wheel Drive EVs**: Tow with rear wheels on dolly
3. **All-Wheel Drive EVs**: Use flatbed only (no dolly towing)
4. Maximum towing speed: 50 km/h (30 mph)
5. Maximum towing distance: 50 km (30 miles)
6. Ensure towed wheels can rotate freely

NEVER:

- Tow with drive wheels on ground (damages motor/inverter)
- Use traditional 2-wheel tow for AWD vehicles
- Exceed manufacturer towing speed/distance limits
- Tow vehicle in "Ready" mode or with HV system active

Post-Collision Towing:

1. Inspect for battery damage before towing
2. Look for coolant leaks or HV component damage
3. If battery damaged: use specialized EV recovery service
4. Transport damaged EVs on flatbed only
5. Inform tow operator of HV system damage

6. Follow hazardous materials transport protocols if required

10.3.5 Long-Term Storage Immobilization

Preparing EV for Storage (>2 weeks):

Battery Preparation:
1. Charge battery to 50-60% SOC (optimal storage level)
2. Do not store at 100% or <20% SOC
3. Disconnect 12V battery to prevent parasitic drain
4. If available, enable "Storage Mode" in vehicle settings

Vehicle Securing:
1. Park on level surface in covered area if possible
2. Engage parking brake
3. Place wheel chocks for added security
4. Inflate tires to maximum pressure (reduces flat-spotting)
5. Close all windows and sunroof

Environmental Protection:
1. Cover vehicle with breathable car cover
2. Ensure adequate ventilation (no sealed garages)
3. Maintain moderate temperature if possible (15-25°C)
4. Protect from direct sunlight and extreme temperatures
5. Ensure dry storage location (no flooding risk)

Monthly Maintenance During Storage:
1. Check tire pressures and inflate if needed
2. Inspect for rodent damage or nesting
3. Check battery SOC if possible (recharge if <40%)
4. Reconnect 12V battery briefly to check systems
5. Move vehicle slightly to prevent tire flat-spots

Returning to Service:
1. Reconnect 12V battery
2. Check tire pressures and adjust to normal
3. Charge HV battery to 80-100%
4. Perform visual inspection of all systems
5. Test drive in safe area before normal use
6. Have vehicle inspected if stored >6 months

10.4 Emergency Services Communication

Description: Critical information to provide to emergency responders for safe and effective EV incident response.

10.4.1 Initial Emergency Call Information

When Calling Emergency Services (Fire/Police/Ambulance):

Essential Information to Provide:

1. **"This is an ELECTRIC VEHICLE emergency"** (state immediately)
2. Exact location (address, GPS coordinates, mile markers)
3. Nature of emergency (fire, collision, electrical hazard, medical)
4. Number of people involved/injured
5. Current hazards (fire, smoke, leaking fluids, electrical sparks)
6. Vehicle make, model, and color
7. Whether occupants are evacuated or trapped

EV-Specific Details:

- High voltage battery location (usually underfloor, state if damaged)
- Visible damage to battery or HV components
- Any warning lights or messages related to HV system
- Presence of fire, smoke, or thermal event
- Chemical odors or unusual sounds
- Coolant leaks (may be pink/orange liquid)

Safety Perimeter Established:

- Distance of safety perimeter from vehicle
- Number of bystanders in area
- Traffic control measures in place
- Wind direction (important for smoke/fumes)

10.4.2 Emergency Responder Information Sheet

Provide to First Responders Upon Arrival:

Vehicle Identification:

- Make: _____
- Model: _____
- Year: _____
- VIN: _____
- Battery Capacity: _____ kWh
- Battery Type: Lithium-ion (specify chemistry if known)

High Voltage System Information:

- HV Battery Location: [Underfloor / Rear / Front / Other]
- Battery Voltage: _____ Volts DC (typically 300-800V)
- Manual Service Disconnect Location: [Specify if known]
- HV Cable Color: ORANGE (high voltage warning)
- Inverter Location: [Front / Rear / Specify]
- Motor Location: [Front / Rear / Both]

Emergency Cut Points:

- 12V Battery Location: [Frunk / Under rear seat / Trunk / Other]
- Avoid cutting orange HV cables
- Safe structural cut points: [Refer to emergency response guide]
- Avoid cutting near: [Battery pack / HV junction box / Inverter]

Current Status:

- Vehicle Power: [ON / OFF / Unknown]
- HV System Status: [Active / Disconnected / Fault / Unknown]
- Visible Damage: [Describe location and severity]
- Fire/Smoke: [Yes / No / Location]
- Coolant Leak: [Yes / No / Color / Location]
- Occupants: [Evacuated / Trapped / Number]

10.4.3 Fire Department Specific Information

⚠ **CRITICAL: EV Battery Fires Require Special Procedures**

Information for Fire Suppression:

Battery Fire Characteristics:

- Lithium-ion battery fires burn at 400-1000°C (750-1800°F)
- May reignite hours or days after initial suppression
- Produce toxic gases (hydrogen fluoride, carbon monoxide, others)
- Require large volumes of water for cooling (10,000-20,000+ liters)
- Thermal runaway can spread cell-to-cell through battery pack
- May burn for hours despite suppression efforts

Recommended Suppression Methods:

1. **Water**: Large volumes for cooling (preferred method)
2. **Foam**: Class A foam with water for surface fires
3. **Dry chemical**: Limited effectiveness on battery fires
4. **Do NOT use**: CO2 (ineffective on lithium-ion fires)

Firefighting Approach:

1. Establish 15-meter (50-foot) safety perimeter

2. Approach from upwind position
3. Use thermal imaging to identify hot spots
4. Apply water to cool battery pack (not just suppress flames)
5. Continue cooling for 30-60 minutes after flames extinguished
6. Monitor for re-ignition for 24-48 hours
7. Consider immersing vehicle in water-filled container if available

Hazardous Materials:

- Toxic smoke: hydrogen fluoride, hydrogen chloride, carbon monoxide
- Electrolyte: flammable organic solvents
- Heavy metals: lithium, cobalt, nickel, manganese
- Corrosive materials: lithium hydroxide (forms with water)
- Electrical hazard: up to 800V DC even when "off"

Personal Protective Equipment:

- Full structural firefighting gear
- SCBA (Self-Contained Breathing Apparatus) mandatory
- Chemical-resistant gloves
- Face shield
- Consider Level A hazmat suit for prolonged exposure

10.4.4 Extrication and Rescue Information

For Trapped Occupants:

Safe Extrication Zones:

- Avoid cutting orange HV cables (typically run along rocker panels)
- Safe cut points: A-pillar (above HV cables), B-pillar (check routing)
- Avoid: Battery pack area (underfloor), HV junction boxes
- Consult Emergency Response Guide (ERG) for specific vehicle

Electrical Hazards:

- Assume HV system energized until proven otherwise
- Use insulated tools rated for >1000V
- Avoid contact with orange cables or components
- Do not cut or penetrate battery pack
- Disable 12V battery if accessible (negative terminal first)

Manual Service Disconnect:

- Location varies by vehicle (often under rear seat or in trunk)
- Requires removal of access panel
- Pull disconnect to de-energize HV system
- Verify HV disconnected using voltage tester
- **Only trained personnel should operate disconnect**

Stabilization:

- Use cribbing to stabilize vehicle before extrication
- Chock wheels to prevent movement
- Disable parking brake if vehicle must be moved
- Monitor for battery swelling or deformation
- Watch for smoke, sparks, or thermal events during extrication

10.4.5 Post-Incident Monitoring and Recovery

24-48 Hour Monitoring Period:

Why Extended Monitoring is Required:

- Battery thermal runaway can occur hours after initial incident
- Damaged cells may slowly heat and reignite
- Chemical reactions continue inside damaged battery
- Re-ignition risk highest in first 24 hours

Monitoring Procedures:

1. **Thermal Monitoring:**
    - **Use thermal imaging camera every 2-4 hours**
    - **Monitor battery pack temperature**
    - **Normal: <40°C; Warning: 40-60°C; Critical: >60°C**
    - **Check for hot spots or temperature increases**

2. **Visual Inspection:**
    - **Look for smoke, steam, or vapors**
    - **Check for battery swelling or deformation**
    - **Inspect for leaking fluids**
    - **Monitor for sparks or electrical arcing**

3. **Atmospheric Monitoring:**
    - **Use gas detectors for hydrogen, carbon monoxide**
    - **Monitor for chemical odors**
    - **Ensure adequate ventilation**
    - **Keep area clear of ignition sources**

Storage of Damaged EV:

- Store outdoors or in well-ventilated area
- Minimum 15 meters from structures or other vehicles
- No enclosed spaces (garage, trailer, etc.)
- Away from combustible materials
- Protected from weather if possible
- Accessible for emergency response

Re-Ignition Response:

- If thermal runaway resumes, repeat suppression procedures
- Apply water to cool battery pack
- Do not attempt to move vehicle during thermal event
- Expand safety perimeter
- Continue monitoring for additional 48 hours after final suppression

Final Disposition:

- Have certified EV technician assess HV system safety
- Battery may require specialized disposal
- Follow hazardous waste protocols
- Document all incident details for insurance and investigation
- Obtain professional clearance before vehicle transport

SECTION 11: COMPREHENSIVE MONITORING SCHEDULE

11.1 Daily Checks (Every Drive)

Pre-Drive Visual Inspection (2-3 minutes):

Exterior Walk-Around:

- Check tire pressures visually (look for obvious low pressure or damage)
- Inspect tires for cuts, bulges, or embedded objects
- Look for fluid leaks under vehicle (coolant, brake fluid)
- Check charging port door closed and locked
- Verify all exterior lights functioning
- Inspect for visible damage from previous day

Interior Pre-Drive Checks:

- Verify dashboard displays no critical warning lights
- Check battery State of Charge (SOC) adequate for planned trip
- Confirm displayed range reasonable for conditions
- Test brake pedal feel (firm, not spongy)
- Verify parking brake releases properly
- Check rearview cameras and sensors functioning

During Drive Monitoring:

- Listen for unusual noises (grinding, clicking, whining)
- Feel for unusual vibrations through steering or pedals
- Monitor regenerative braking consistency
- Watch for warning lights or messages

- Note any changes in acceleration or power delivery
- Observe energy consumption rate (kWh/km or kWh/mile)

Post-Drive Checks:

- Note any warning messages that appeared during drive
- Check for unusual odors (burning, chemical, sweet coolant smell)
- Verify vehicle shuts down properly
- Look under vehicle for new leaks after parking
- Note battery temperature if displayed (should cool within 30 min)

11.2 Weekly Checks (Every 7 Days or 500 km)

Battery System (5 minutes):

- Check battery SOC trend at similar charge levels
- Monitor charging completion time for consistency
- Verify battery temperature normal after charging (should be <40°C)
- Check for battery-related warning messages in vehicle logs
- Document displayed range at 100% SOC (track degradation)
- Verify battery cooling fan operates during charging (listen for it)

Tire System (10 minutes):

- **Measure tire pressures with gauge (all 4 tires + spare)**
    - Front tires: _____ PSI (spec: _____ PSI)
    - Rear tires: _____ PSI (spec: _____ PSI)
    - Spare: _____ PSI
  - Inflate to manufacturer specification if needed
  - Inspect tread depth (minimum 4mm / 5/32")
  - Check for uneven wear patterns (edge vs. center wear)
  - Look for sidewall damage, bulges, or cracks
  - Remove stones or debris from tread grooves
  - Verify TPMS sensors reporting correctly

Charging System (5 minutes):

- Inspect charging cable for damage or wear
- Clean charging port contacts with dry cloth
- Check charging port for debris or corrosion
- Verify charging starts within 30 seconds of plugging in
- Monitor actual charging power vs. expected (within 10%)
- Listen for unusual sounds from on-board charger

Fluid Levels (5 minutes):

- Check coolant level in reservoir (when cold)
- Inspect windshield washer fluid level
- Check brake fluid level (should be between MIN/MAX)
- Look for signs of leaks around fluid reservoirs

- Verify no coolant stains on driveway/parking spot

12V Battery (2 minutes):

- Check 12V battery voltage if accessible (should be >12.4V when off)
- Verify no corrosion on 12V battery terminals
- Test that all accessories function properly (lights, wipers, etc.)
- Check for "12V Battery Low" warnings

11.3 Monthly Checks (Every 30 Days or 2,000 km)

Comprehensive Battery Assessment (15 minutes):

- **Perform capacity test:**
    1. Charge to 100% SOC
    2. Note displayed range: _____ km/miles
    3. Drive until 10% SOC
    4. Record kWh consumed: _____ kWh
    5. Compare to previous tests and manufacturer specs
- Check cell voltage balance (if accessible via app/OBD):
    - Maximum cell voltage: _____ V
    - Minimum cell voltage: _____ V
    - Deviation: _____ mV (should be <50mV)
- Review battery temperature trends during charging
- Document any changes in charging behavior
- Perform balancing charge (100% SOC, hold 2-3 hours)

Motor and Drivetrain (10 minutes):

- Listen for motor noise during consistent acceleration test
- Check motor temperature after 30-minute highway drive
- Test acceleration performance (0-60 mph time if safe)
- Verify no vibration at various speeds (40, 60, 80 km/h)
- Check for drivetrain noises (clicking, grinding, whining)
- Inspect motor compartment for coolant leaks
- Verify coolant level adequate

Inverter and Power Electronics (5 minutes):

- Monitor inverter temperature during normal driving
- Test regenerative braking at various speeds
- Check for high-pitched whining from inverter area
- Verify no "Power Limited" warnings during normal driving
- Review energy consumption efficiency vs. baseline

Thermal Management (10 minutes):

- Check coolant level in all reservoirs (when cold)
- Inspect coolant color and clarity (should be clear, proper color)
- Look for coolant leaks under vehicle and around components

- Test cooling fan operation (should activate during charging)
- Listen for cooling pump noise (should be quiet hum)
- Verify pre-conditioning function works properly
- Check cabin HVAC performance (heating and cooling)

Braking System (5 minutes):

- Test brake pedal feel (should be firm, not spongy)
- Check brake fluid level (between MIN/MAX marks)
- Listen for brake noise (squealing, grinding)
- Verify regenerative braking consistent and smooth
- Test parking brake holds vehicle on incline
- Inspect brake pads through wheels (if visible, >3mm remaining)

High Voltage Safety (5 minutes):

- Check for any HV system warning lights or messages
- Verify no tingling or shock sensations when touching vehicle
- Inspect HV cable routing for damage (orange cables)
- Check interlock system status (if accessible)
- Verify no unusual electrical smells
- Review insulation resistance if accessible via diagnostics

Exterior and Interior (10 minutes):

- Wash vehicle and inspect for damage
- Check all lights functioning (headlights, taillights, turn signals)
- Test windshield wipers and washers
- Inspect wiper blades for wear
- Check cabin air filter condition (replace if needed)
- Test all door locks and latches
- Verify charging port door operates smoothly
- Clean charging port contacts

Software and Diagnostics (10 minutes):

- Check for available software updates (OTA or dealer)
- Review diagnostic trouble codes (DTCs) using OBD-II reader or app
- Document any warning messages from past month
- Verify vehicle connectivity (cellular, WiFi) functioning
- Check mobile app functions properly
- Review energy consumption statistics
- Export and save vehicle data logs if available

11.4 Quarterly Checks (Every 3 Months or 6,000 km)

Detailed Battery Analysis (30 minutes):

- **Full capacity test with detailed documentation:**
  - Starting SOC: _____ %

    - Ending SOC: _____ %
    - Distance driven: _____ km/miles
    - Energy consumed: _____ kWh
    - Average efficiency: _____ kWh/km or kWh/mile
    - Temperature range: _____ to _____ °C
    - Driving conditions: [Highway/City/Mixed]
- Compare capacity to previous quarters (track degradation rate)
- Review cell voltage balance trends over past 3 months
- Analyze charging time trends (20-80% charge duration)
- Document any battery-related warnings or events
- Check battery warranty coverage and remaining capacity guarantee

Tire Rotation and Inspection (45 minutes):

- **Rotate tires per manufacturer schedule:**
    - Front-to-rear or cross-pattern as specified
    - Document tire positions before and after rotation
        - **Measure tread depth at multiple points per tire:**
            - LF outer: _____ mm, center: _____ mm, inner: _____ mm
            - RF outer: _____ mm, center: _____ mm, inner: _____ mm
            - LR outer: _____ mm, center: _____ mm, inner: _____ mm
            - RR outer: _____ mm, center: _____ mm, inner: _____ mm
- Calculate wear rate (mm per 1000 km)
- Check wheel balance (if vibration present)
- Inspect wheels for bends, cracks, or damage
- Check lug nut torque (use torque wrench)
- Inspect valve stems for cracks or damage
- Consider wheel alignment if uneven wear detected

Cooling System Service (30 minutes):

- **Detailed coolant inspection:**
    - Check level in all reservoirs
    - Inspect color (should be pink/orange, not brown)
    - Check for contamination or debris
    - Test pH if test strips available (should be 7.5-11.0)
- Pressure test cooling system (if equipment available)
- Inspect all coolant hoses for cracks, bulges, or leaks
- Check hose clamps for tightness
- Inspect radiator fins for blockage or damage
- Clean radiator exterior (low-pressure water)
- Test cooling fan operation at various speeds
- Listen for cooling pump noise or cavitation
- Check coolant temperature sensors functioning

Brake System Detailed Inspection (20 minutes):

- Measure brake pad thickness (all four corners)
    - Front pads: _____ mm remaining (minimum 3mm)
    - Rear pads: _____ mm remaining (minimum 3mm)
- Inspect brake rotors for scoring, warping, or cracks

- Check brake fluid condition (should be clear, not dark)
- Test brake pedal travel and firmness
- Verify ABS and traction control functioning
- Check parking brake adjustment and hold strength
- Inspect brake lines for corrosion or damage
- Verify brake wear sensors functioning (if equipped)

High Voltage System Inspection (20 minutes):

- **Professional HV system health check (dealer or certified shop):**
    - Insulation resistance test: _____ MΩ (should be >100 MΩ)
    - Leakage current: _____ mA (should be <0.5 mA)
    - HV battery voltage: _____ V
    - Cell voltage range: _____ to _____ V
    - Interlock system test: [Pass/Fail]
    - Ground fault test: [Pass/Fail]
  - Inspect HV cables for damage or chafing
  - Check HV connectors for corrosion
  - Verify HV contactors operating properly
  - Test manual service disconnect function
  - Review HV system diagnostic codes

Suspension and Steering (15 minutes):

- Inspect shock absorbers for leaks
- Check suspension bushings for wear or cracking
- Test steering for play or looseness
- Verify power steering functioning smoothly
- Inspect CV boots for tears or damage
- Check ball joints and tie rod ends
- Look for signs of suspension damage or wear
- Test ride quality over bumps (should be controlled)

Comprehensive Vehicle Scan (30 minutes):

- **Connect OBD-II scanner or professional diagnostic tool:**
  - Scan all modules for DTCs
  - Review freeze frame data
  - Check sensor readings (temperatures, voltages, pressures)
  - Verify all systems communicating properly
- Document all diagnostic codes found:
  - Code: _____ Description: _____ Status: _____
  - Code: _____ Description: _____ Status: _____
- Clear temporary codes after verification
- Export diagnostic report for records
- Review with service advisor if concerning codes found

11.5 Annual Checks (Every 12 Months or 20,000 km)

Complete Professional Inspection: Schedule comprehensive service at authorized dealer or certified EV service center including:

Battery System:

- Professional battery health assessment
- State of Health (SOH) measurement and documentation
- Cell balance verification across entire pack
- Thermal management system performance test
- BMS software update if available
- Battery warranty review and documentation

Electrical Systems:

- HV system comprehensive safety inspection
- Insulation resistance testing
- Ground fault detection system verification
- 12V battery load test and replacement if needed
- All electrical connections inspection and cleaning
- Software updates for all vehicle modules

Mechanical Systems:

- Motor bearing inspection and lubrication if required
- Drivetrain inspection (gearbox, axles, CV joints)
- Suspension comprehensive inspection
- Steering system inspection and alignment check
- Wheel alignment (four-wheel alignment recommended)
- Brake system complete inspection and fluid replacement

Fluid Services:

- Coolant condition test and replacement if needed (typically 5-8 years)
- Brake fluid replacement (every 2-3 years)
- Gearbox oil check/replacement if applicable
- Windshield washer fluid top-up

Safety Systems:

- Airbag system diagnostic check
- Seatbelt inspection and pretensioner verification
- Crash sensor system verification
- Emergency response system test
- High voltage interlock system comprehensive test

Cabin and Comfort:

- Cabin air filter replacement
- HVAC system performance test and cleaning
- Climate control refrigerant check

- All interior controls and features verification

Documentation and Records:

- Update service history in vehicle records
- Document all measurements and test results
- Photograph any wear or damage for future reference
- Update maintenance schedule for next service
- Review warranty coverage and expiration dates
- Export and archive all diagnostic data

SECTION 12: GLOSSARY OF TECHNICAL TERMS

AC (Alternating Current): Electrical current that reverses direction periodically. Used by the motor and supplied by charging stations.
Balancing (Battery): Process where BMS equalizes voltage across all cells by discharging higher-voltage cells or charging lower-voltage cells.
BMS (Battery Management System): Electronic controller that monitors and manages battery pack, including voltage, temperature, and state of charge.
Cell: Individual lithium-ion battery unit, typically 3.6-4.2V. EVs contain hundreds to thousands of cells.
DC (Direct Current): Electrical current flowing in one direction. Battery stores DC power; fast chargers supply DC directly to battery.
DC-DC Converter: Device that converts high-voltage DC from traction battery to low-voltage DC (12V) for vehicle accessories.
DTC (Diagnostic Trouble Code): Standardized code indicating a specific fault detected by vehicle systems.
EVSE (Electric Vehicle Supply Equipment): Charging station that supplies electrical power to charge EV battery.
Ground Fault: Electrical current leaking from high-voltage system to vehicle chassis (ground), creating shock hazard.
HV (High Voltage): Electrical system operating above 60V DC or 30V AC. EV traction systems typically 300-800V.
IGBT (Insulated Gate Bipolar Transistor): Power semiconductor switch used in inverters to convert DC to AC.
Insulation Resistance: Measure of electrical resistance between HV system and vehicle chassis. High resistance (>100 MΩ) indicates good insulation.
Interlock: Safety switch that automatically disconnects HV system when service panels are opened or in crash.
Inverter: Power electronics that convert DC battery power to AC motor power and control motor speed/torque.
**kWh (Kilowatt-
hour):** Unit of energy. Battery capacity measured in kWh (e.g., 75 kWh battery). Energy consumption measured in kWh/100km or kWh/mile.
Leakage Current: Small electrical current flowing where it shouldn't, such as from HV system to chassis ground.

Module: Group of battery cells connected together, typically 12-24 cells per module. Multiple modules form the battery pack.
OBC (On-Board Charger): Device that converts AC power from charging station to DC power for battery charging.
OBD-II (On-Board Diagnostics): Standardized vehicle diagnostic system accessible through 16-pin connector, used to read fault codes and sensor data.
Phase Current: Electrical current in each of the three phases (A, B, C) supplying AC power to the motor.
PMSM (Permanent Magnet Synchronous Motor): Type of electric motor using permanent magnets in rotor, common in EVs.
Power Factor: Ratio of real power to apparent power in AC systems. Higher is better (>0.95 optimal).
PWM (Pulse Width Modulation): Switching technique used by inverter to create variable AC voltage from DC battery.
Regenerative Braking (Regen): Process of converting vehicle kinetic energy back to electrical energy during deceleration, charging the battery.
Rotor: Rotating part of electric motor, contains magnets (PMSM) or conductors (induction motor).
SOC (State of Charge): Battery charge level expressed as percentage (0-100%). Similar to fuel gauge in conventional vehicles.
SOH (State of Health): Battery capacity as percentage of original capacity. New battery = 100% SOH; degraded battery = <100% SOH.
Stator: Stationary part of electric motor containing electromagnetic windings that create rotating magnetic field.
Thermal Runaway: Self-accelerating overheating of battery cells, potentially leading to fire. Caused by internal short circuit, overheating, or damage.
THD (Total Harmonic Distortion): Measure of electrical waveform distortion. Lower THD indicates cleaner power.
TPMS (Tire Pressure Monitoring System): System using sensors to monitor tire pressure and temperature, alerting driver to problems.
Traction Battery: Main high-voltage battery pack that powers the electric motor (also called HV battery or propulsion battery).
12V Battery: Auxiliary low-voltage battery that powers vehicle accessories, computers, and controls. Similar to battery in conventional vehicles.

SECTION 13: QUICK REFERENCE EMERGENCY GUIDE

⚠ CRITICAL WARNINGS - IMMEDIATE ACTION REQUIRED

STOP DRIVING IMMEDIATELY IF:

- Battery temperature >50°C (122°F)
- Smoke or fire from any component

- "High Voltage System Fault" warning
- Burning smell or unusual chemical odor
- Rapid pressure loss in tire (>2 PSI/minute)
- Severe vibration or grinding noises
- Coolant leak (pink/orange liquid)
- Insulation fault or ground fault warning
- Any shock or tingling sensation from vehicle
- Sudden loss of power or braking ability

EVACUATE VEHICLE IF:

- Fire or smoke visible
- Strong chemical smell
- Battery swelling or deformation visible
- Popping or hissing sounds from battery area
- After collision with battery damage suspected
- Persistent electrical arcing or sparks

Emergency Contact Numbers

Keep These Numbers Accessible:

- Emergency Services: **911** (or local emergency number)
- Roadside Assistance: _____
- Vehicle Manufacturer Hotline: _____
- Insurance Company: _____
- Authorized Service Center: _____
- Towing Service (EV-capable): _____

Quick Diagnostic Reference

Warning Light → Meaning → Action Required

🔋 **Red Battery Warning**

- Critical battery system fault
- **STOP IMMEDIATELY** - Do not drive
- Call roadside assistance

⚡ **Red High Voltage Warning**

- HV system fault or safety issue
- **STOP IMMEDIATELY** - Do not drive
- Evacuate if smoke/smell present

🌡️ **Red Temperature Warning**

- Component overheating (battery/motor/inverter)
- **STOP IMMEDIATELY** - Allow cooling
- Do not restart until temperature normal

🔧 **Red Drivetrain Warning**

- Motor or inverter critical fault

- **STOP IMMEDIATELY** - Do not drive
- Call service center

⚠ **Amber Battery Warning**

- Battery system monitoring required
- **Safe to drive** - Schedule service within 1 week
- Avoid fast charging until diagnosed

⚠ **Amber Tire Pressure Warning**

- Low tire pressure detected
- **Check immediately** - Inflate tires
- Safe to drive short distance to air pump

⚠ **Amber Brake Warning**

- Brake system issue or low fluid
- **Check immediately** - Inspect brake fluid
- Avoid aggressive driving until resolved

⚠ **Amber Cooling System Warning**

- Cooling system issue
- **Check coolant level** - Top up if low
- Reduce power demands, schedule service

Emergency Shutdown Procedure

If Vehicle Must Be Disabled Immediately:

1. Bring to controlled stop
    - Apply brakes firmly
    - Shift to PARK
    - Engage parking brake
2. Turn off vehicle
    - Press and hold power button
    - Verify "Ready" light OFF
3. Disconnect 12V battery (only if safe, no fire/smoke)
    - Open hood/frunk
    - Disconnect NEGATIVE (-) terminal first
    - Disconnect POSITIVE (+) terminal second
4. Evacuate and establish safety perimeter
    - Move 15 meters (50 feet) away
    - Keep others away
    - Call emergency services
5. Do NOT:
    - Open HV service panels
    - Touch orange cables
    - Attempt to move vehicle
    - Re-enter vehicle if hazard present

Thermal Event Response (Battery Fire)

If Battery Fire or Thermal Runaway Occurs:

1. EVACUATE IMMEDIATELY
    - Get all occupants out
    - Move 15+ meters away
    - Do NOT attempt to fight fire
2. CALL 911
    - State "ELECTRIC VEHICLE FIRE"
    - Provide exact location
    - Report battery capacity if known
    - State if occupants safe
3. ESTABLISH SAFETY PERIMETER
    - Keep everyone 15 meters away
    - Warn approaching vehicles
    - Position upwind of smoke
    - Do not re-approach vehicle
4. PROVIDE INFORMATION TO RESPONDERS
    - Vehicle make, model, year
    - Battery location (underfloor)
    - Visible damage location
    - Duration of thermal event
5. MONITOR FOR 24-48 HOURS
    - Battery may reignite
    - Continue safety perimeter
    - Store outdoors only
    - Professional assessment required

Collision Response Checklist

Immediately After Collision:

- Check for injuries
- Assess vehicle stability
- Look for fire/smoke/leaks
- Turn off vehicle if possible
- Activate hazard lights
- Call 911 if injuries or hazards
- Establish safety perimeter if HV damage suspected
- Document damage with photos
- Inform responders vehicle is electric
- Do NOT disconnect 12V battery (preserves crash data)
- Wait for professional HV safety assessment
- Arrange flatbed towing only

Towing Quick Reference

ALWAYS USE FLATBED TOW TRUCK

✅ **Correct:**

- All four wheels off ground
- Flatbed or wheel-lift tow truck
- Parking brake released before loading
- Secure with wheel straps

❌ **NEVER:**

- Tow with drive wheels on ground
- Use traditional 2-wheel tow
- Tow AWD vehicles on dolly
- Exceed 50 km/h or 50 km if dollying

After Collision:

- Inspect for battery damage first
- Use flatbed only if HV damage
- Inform tow operator of damage
- Follow hazmat protocols if required

Temperature Thresholds Quick Reference

Component	Normal	Warning	Critical
Battery	15-35°C	35-45°C	>45°C
Motor	60-90°C	90-110°C	>110°C
Inverter	50-75°C	75-90°C	>90°C
OBC	40-65°C	65-80°C	>80°C
Tire	Ambient +10-30°C	+30-50°C	>+50°C

If Critical Temperature Reached:

- Stop vehicle immediately
- Allow cooling (15-30 minutes minimum)
- Do not restart until temperature normal
- Investigate cause before continued driving
- Schedule service if temperatures persist

Pressure and Voltage Quick Reference

Tire Pressure:

- Normal: 32-36 PSI (check door placard)
- Warning: 28-32 PSI
- Critical: <28 PSI
- **Action:** Inflate immediately if <30 PSI

Battery Voltage:

- Normal: 300-420V (model-specific)
- Warning: 280-300V or 420-450V
- Critical: <280V or >450V
- **Action:** Service required if outside normal range

12V Battery:

- Normal: 12.6-14.4V
- Warning: 12.0-12.6V
- Critical: <12.0V
- **Action:** Charge or replace if <12.4V when off

Insulation Resistance:

- Normal: >500 Ω/V (>200 MΩ @ 400V)
- Warning: 100-500 Ω/V
- Critical: <100 Ω/V
- **Action:** DO NOT DRIVE if <100 Ω/V

Service Timing Quick Reference

IMMEDIATE (Stop Driving):

- Any red warning light
- Fire, smoke, or burning smell
- Battery temp >50°C
- Tire pressure <25 PSI
- Insulation fault
- Ground fault
- Severe vibration or noise
- Coolant leak

WITHIN 24 HOURS:

- Amber HV system warning
- Significant performance loss
- Charging failure
- Brake system warning
- Coolant level low
- Tire damage after impact

WITHIN 1 WEEK:

- Persistent amber warnings
- Battery temp frequently >40°C
- Charging rate reduced >15%
- Motor/inverter temp elevated
- Unusual noises developing
- Efficiency loss >5%

WITHIN 1 MONTH:

- Minor amber warnings
- Gradual performance changes
- Efficiency loss 3-5%
- Coolant discoloration
- Minor wear patterns

NEXT SCHEDULED SERVICE:

- Preventive maintenance items
- Minor efficiency changes
- Routine inspections
- Software updates

Owner-Serviceable vs. Professional Service

✅ **OWNER CAN PERFORM:**

- Tire pressure checks and inflation
- Visual inspections
- Fluid level checks (coolant, brake, washer)
- Cleaning charging port
- 12V battery disconnect (if trained)
- Software updates (OTA)
- Basic diagnostics (OBD-II scanner)
- Tire rotation (if trained and equipped)

❌ **REQUIRES PROFESSIONAL SERVICE:**

- Any HV system work
- Battery pack service
- Motor or inverter service
- Coolant system service
- Brake system service
- Wheel alignment
- HV cable or connector service
- Insulation testing
- Manual service disconnect operation
- Post-collision HV inspection
- Any work requiring HV certification

Emergency Equipment Recommended

Keep In Vehicle:

- Tire pressure gauge (digital recommended)
- Portable tire inflator (12V or battery-powered)
- Tire repair kit
- Wheel chocks
- Warning triangles or road flares
- Fire extinguisher (Class ABC, 5+ lbs)
- First aid kit
- Flashlight with extra batteries
- Emergency contact list
- Vehicle emergency response guide
- Insulated gloves (if trained for 12V disconnect)
- Reflective vest
- Bottled water and non-perishable snacks

In Home Garage:

- OBD-II scanner or diagnostic app
- Torque wrench (for lug nuts)
- Jack and jack stands (if tire changes performed)
- Multimeter (for 12V battery testing)
- Coolant test strips (pH testing)
- Service manual or emergency response guide
- Tire tread depth gauge

Manual Version: 1.0
Last Updated: 2024
For Technical Support: Contact your vehicle manufacturer or authorized service center
⚠️ **DISCLAIMER:** This manual provides general guidance for EV preventive maintenance. Always consult your vehicle's specific owner's manual and follow manufacturer recommendations. High-voltage system work must only be performed by certified technicians with proper training and equipment.